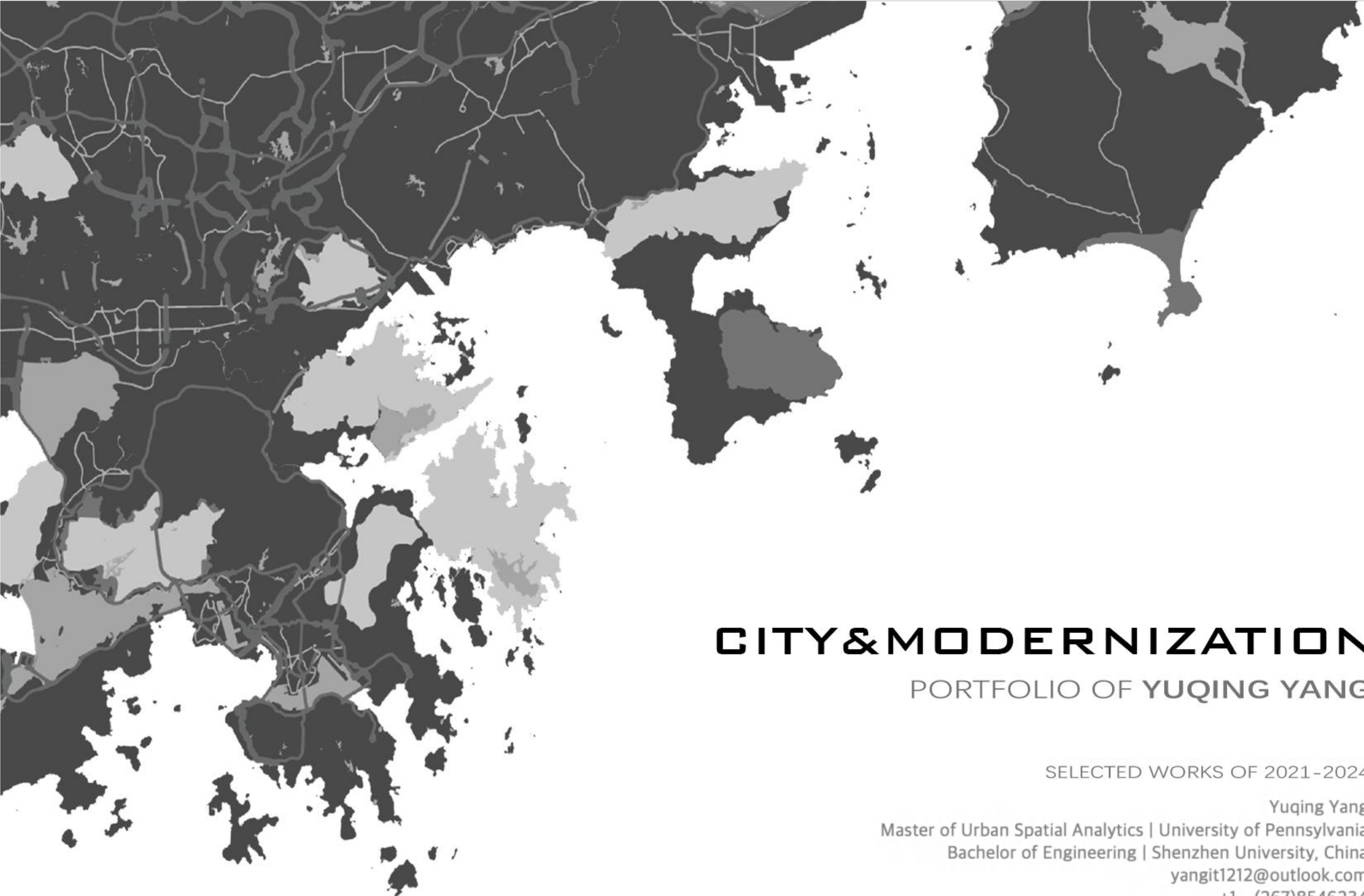

CITY&MODERNIZATION

PORTFOLIO OF **YUQING YANG**

SELECTED WORKS OF 2021-2024

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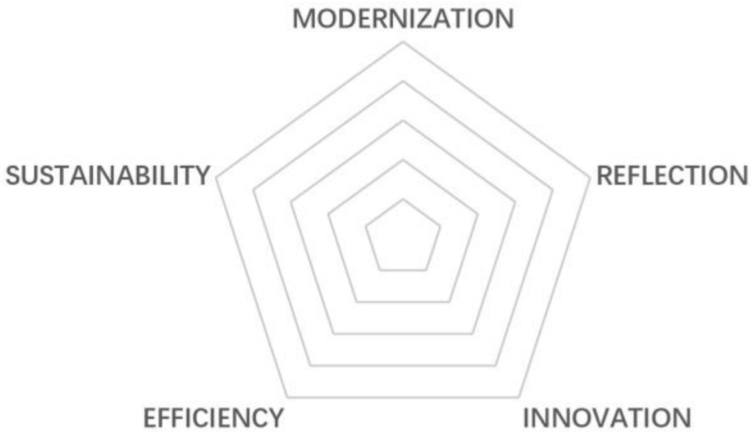


PERSONAL MONOLOGUE

During my five years of undergraduate study, my understanding of cities has expanded from small-scale individual buildings to large-scale regional planning. In this process, I also experienced spatial design based on sensibility and feeling as well as urban research based on logic and rationality.

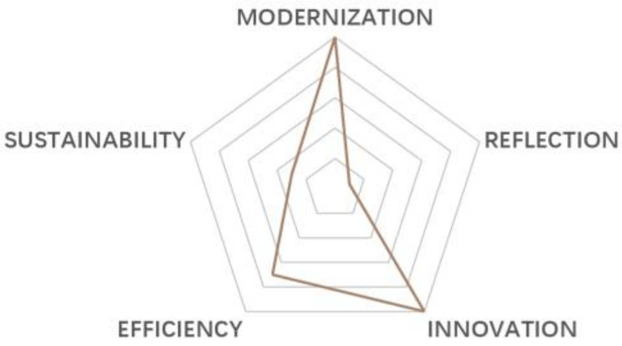
I am deeply aware that urban planning needs to be fully integrated with **modernization**, and urban development cannot be separated from **innovation, reflection, sustainability and efficiency**. So, I want to use modern technology to help cities grow **more flexible, more visual, more efficient and more sustainable.**

This portfolio fully shows my thoughts and pursuits. Through the four projects presented in the portfolio, I hope to explore a more rational and scientific approach to urban research.

CONTENTS

01 "EXTRATERRESTRIAL"

Suitability of the Location of UAVs' Take-off and Landing Points in the Field of Real-time Delivery



Type: Academic work | *Comprehensive investigation of urban and rural society*
Instructor: Wenxiu Gao
Date: 09/2023-01/2024
Location: Shenzhen, China
Group work： Yuqing Yang, Qiaolin Quan, Yongqi Li, Zihao Xu
Awards: Nomination Awards | *Wupencity 2024 International Competition on Urban Sustainability Reports*

Urban planning needs innovation. While scientific and technological progress brings more convenient human life, advanced equipment and advanced technology will certainly have an impact on urban form. Therefore, cities must face the rapid development of modern society with a flexible, innovative and inclusive attitude.

At present, UAV distribution in the field of instant distribution has reached a high degree of maturity. In Shenzhen, a number of routes have also been in stable operation. Through the spatial analysis of the flight route and the landing point, the **core factors of the UAV takeoff and landing point location** were extracted and condensed, and the **formula of the suitability** of the landing point was calculated. At the same time, the potential plots and potential points are screened out in the Nanshan district, and the suitability formula is used to verify the potential points, which proves the establishment of the model and establishes the key factors affecting the deployment of take-off and landing points. Finally, **policy suggestions and promotion suggestions** are put forward through the suitability study of UAV take-off and landing point location in the field of instant distribution.

1 Introduction

1.1 Background

Low altitude usually refers to the airspace within 1000 meters of the vertical distance from the ground plane directly below, and can be extended to the airspace within 3000 meters according to the characteristics of different regions and actual needs. The low-altitude economy is based on the low altitude airspace and driven by the low-altitude flight activities of various manned and unmanned aircraft. It is widely reflected in various industrial forms. In February 2021, the concept of "low-altitude economy" was written into the national plan for the first time.

Unmanned Aerial Vehicles (UAV) is an unmanned aircraft operated by radio remote control equipment and its own program control device. The UAV industry has gradually become the leading industry of low altitude economy. At present, industrial grade Uavs for civilian use have been widely used in agriculture, forestry, plant protection, power line patrol, border patrol, forest fire prevention, **logistics and distribution.**

1.2 Developing Status

In 2022, the global low-altitude economy market scale has reached the level of $100 billion. Among them, the UAV and aircraft market accounted for about 70%. In China, there are **15,000** domestic UAV operating enterprises with an annual output value of 117 billion yuan.

In 2023, the annual output value of Shenzhen's low-altitude economy has exceeded 90 billion yuan, which is 20% higher than that of last year. 77 new UAV routes were opened, while 73 new UAV take-off station were built , and 600,000 cargo UAV flights were completed, **ranking first in the country**, and consumer drones accounted for 70% of the global market share.

1.3 Application Scenario

Emergency Distribution	Express Logistics	Instant Deliveries
Medical cold chain, biological samples, etc. Strong immediacy, High value goods.	Remote areas; Fresh delivery; Long mileage, Large demand, Heavy loads.	Food, daily necessities, etc. Short mileage, Large demand, Strong immediacy.

1.4 Research Question

In addition to basic functions such as auxiliary transportation, tourism aerial photography, commercial performance, as one of the strategic emerging industries, UAV logistics has developed rapidly in the past decade. China has made progress in UAV technology, products, applications and markets, but **the normalization of routes are difficult.**

Core technology issues?	Talent shortage issues?
Is the industry supervision insufficient?	Is the spatial layout unreasonable?

UAV delivery routes in Shenzhen

Drones for consumer use in Shenzhen account for **70% of the global market share**, becoming the focus city of the national and even global drone industry. In order to help Shenzhen achieve the goal of "terminal 3km, 15-minute distribution circle", Meituan company has opened 13 regular operation routes.

1.5 Research Purpose

- To explore what factors should be considered in the arrangement of UAV take-off and landing points from the perspective of **crowd needs** and **spatial characteristics.**
- To provide a reference basis for evaluating **the suitability of the land plots** for the future UAVs.
- To help Shenzhen hit the **low altitude economy "first city".**
- To ensure the **public interest** in spatial layout. To explore the connection between new technologies and the updating of people's life scenes.

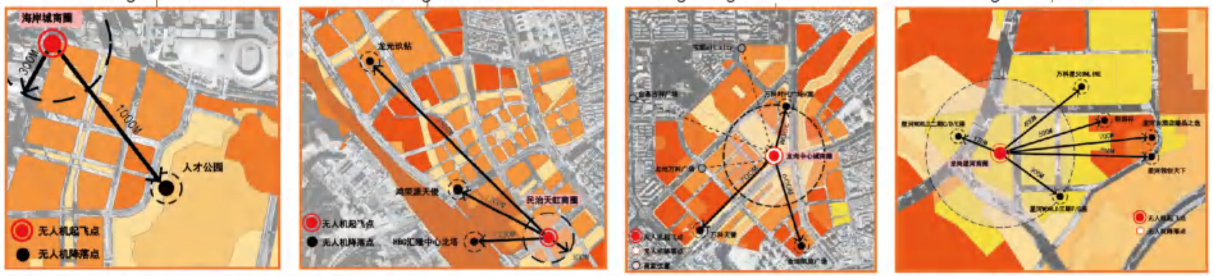
1.6 Study Area

Current UAV delivery routes in Shenzhen

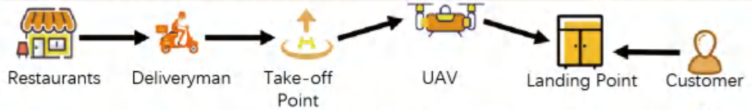
At present, Shenzhen has opened a number of normal operation routes, including residential areas, office areas and scenic spots.

Industrial Land
Green Space and Open Space
Land Water Area
Residential Land
Transportation Land
Take-off Point
Landing Point

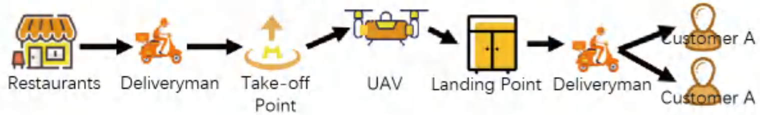
Hai' an cheng business circle | Tianhong business circle | Longcheng business circle | Xinghe business circle



Take-off Point	Landing Point	Number	Take-off Point	Landing Point	Number
Longcheng Business Circle	万科时代广场V寓	A1	Xinghe Business Circle	星河吉酒店臻品之选	A3
	金地凯旋广场	A2		银湖谷	A4
	万科天誉	A5		星河WORLD F/G 座	B4
Tianhong Business Circle	鸿荣源天俊	B1		星河WORLD C/D 座	B5
	龙光玖钻	B2		星河创领天下	B6
	HBC汇隆中心北塔	B3		万科星火ONLINE	B7
Note: The landing points of Class A routes are set in residential areas; Class B is set in the office area; Class C is set in scenic spots			Hai' an cheng Business Circle	人才公园	C1



Routes	A3	B5 B2 B6	C1



Routes	A1 A2 A4 A5	B1 B2 B3 B4 B7

2 Field Research

2.1 Data Category

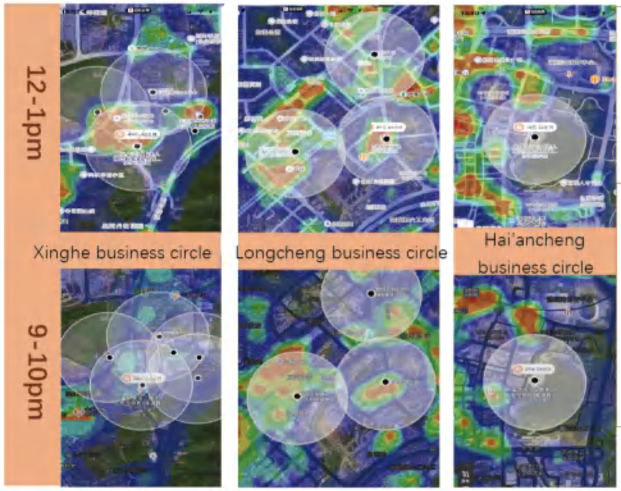
a Location and Land use		b Potential user base			
a1 Type of land use	a2 District function(within 300m)	b1 Number of potential users(within 300m)	b2 Crowd size heat map	b3 Building density	b4 Plot ratio near landing points

c Service point accessibility				d Visibility				e	f
c1 Number of shops nearby (10-50 m)	c2 Number of shops nearby (within 300m)	c3 Accessibility to surrounding shops	c4 Accessibility between restaurants and landing points	d1 Spatial location	d2 Closet distance to the surrounding building	d3 Closet distance to the entrance of surrounding building	d4 Closet distance to the surrounding leisure seat	e1 Degree of functional complexity	f1 Terminal distribution form

2.2 Research Results

Routes with landing points located in **office areas with high density and high floor area ratio** have the most potential users, followed by those landing points located in residential areas with high floor area ratio. The office and scenic areas with low density and low volume ratio have the least number of potential people.

b2 Crowd size heat map

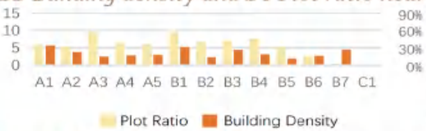


The airdrop cabinets arranged near office buildings and residential areas are usually located in or on **the edge of populated areas** so as to serve more users in a regular and large-scale manner

The location of the airdrop cabinet arranged in the scenic spot reduces the consideration of the population density, **demonstration and showing brand image**may be more important. Therefore, conspicuous location in the scenic spot can be selected

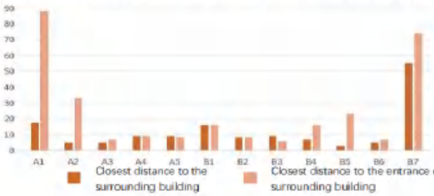
500m ● Airdrop cabinet (Landing point)

b3 Building density and b4 Plot ratio near landing points



The plot ratio of the landing points in the residential area and office area are both high, but the **building density varies greatly.**

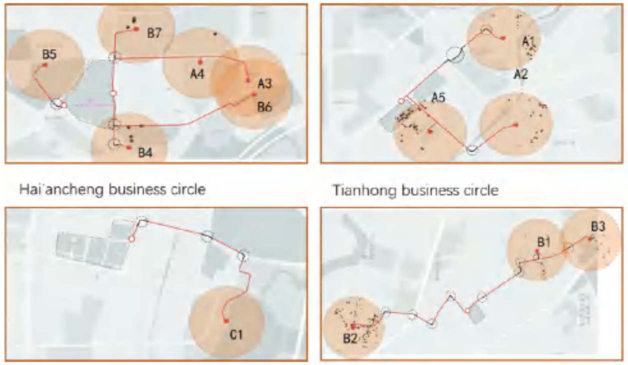
d2 d3 Closest distance to(the entrance of)surrounding building



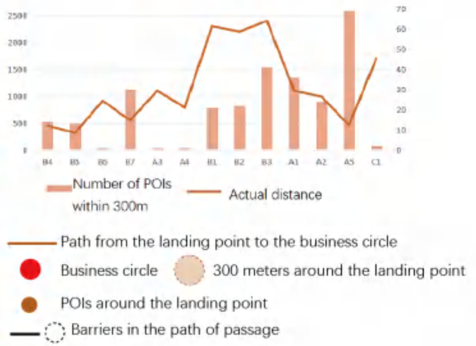
The distance of the airdrop cabinet at the landing point is basically **less than 20m** from the surrounding buildings. In the case that the airdrop cabinet is too far away from the entrance and exit of the building, **another deliveryman** will be equipped to serve the guests to ensure the convenience of taking food

c Service point accessibility

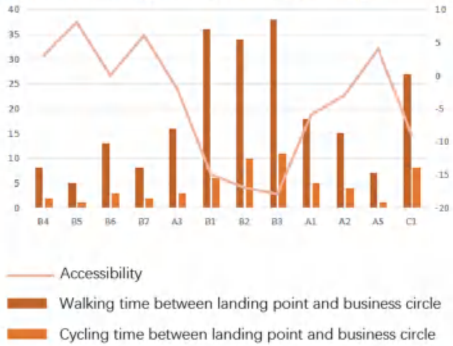
Xinghe business circle / Longcheng business circle / Hai'ancheng business circle / Tianhong business circle



c2 Number of shops nearby (within 300m)



c3 Accessibility to surrounding shops

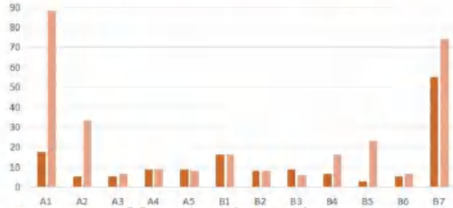


Accessibility evaluation of each landing point to the business circle

B4	B5	B6	B7	A3	A4	B1	B2	B3	A1	A2	A5	C1
3	8	0	6	-2	-1	-15	-17	-18	-6	-3	4	-9

On the whole, the road accessibility from the landing point to the Tianhong business circle(B1, B2, B3) is poor and the number of POIs around the landing point is small, so it is more suitable to apply UAV distribution here. According to the walking distance data from the landing point to the business district, the walking time from all the landing points to the business district is more than 5min. It can be said that the UAV distribution makes up for the vacancy of the **5min life circle** to a certain extent. And there are no POI points around some landing points, which proves that it needs UAV distribution to make up for the catering service capacity in the area.

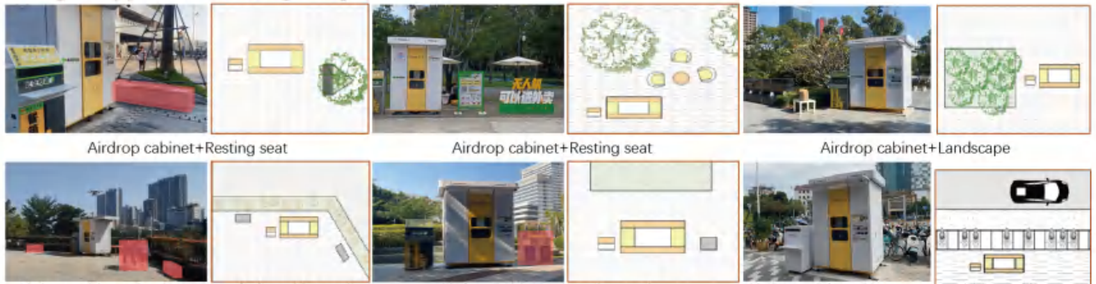
d2 d3 Closest distance to(the entrance of)surrounding building



The distance of the airdrop cabinet at the landing point is basically **less than 20m** from the surrounding buildings. In the case that the airdrop cabinet is too far away from the entrance and exit of the building, **another deliveryman** will be equipped to serve the guests to ensure the convenience of taking food.

Closest distance to the surrounding building
Closest distance to the entrance of surrounding building

e1 Degree of functional complexity



Airdrop cabinet+Resting seat | Airdrop cabinet+Resting seat | Airdrop cabinet+Landscape

Airdrop cabinet+Resting seat、clothes airing pole | Airdrop cabinet+dumping stations | Airdrop cabinet+Non-motor vehicle parking area

According to different usage scenarios, the airdrop cabinet can combine different functions. For example, a low-altitude economic science exhibition can be set up next to the airdrop cabinet in the park.

Whether there is terminal delivery

A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1
○	○	×	○	○	○	○	○	○	×	×	○	×

Take-off point situation

Take-off point	Xinghe business circle	Longcheng business circle	Tianhong business circle	Hai' an cheng business circle
Number of merchants	54	67	44	93
Basic takeout orders of merchants	55800	45700	33400	67468
Takeoff scene	Roof of the mall	Roof of the mall	A corner of the park	Second floor platform
Occupied area	Around 300m²	Around 36m²	Around 50m²	Around 40m²
Number of UAV delivery merchants	17	6	12	4

3 Data Modeling

3.1 Methods

Analytic Hierarchy Process

The elements related to decision making are decomposed into multiple levels such as goals, criteria and schemes, and qualitative and quantitative analysis is carried out based on these levels.

The steps mainly include three: the construction of hierarchical structure model, the construction of judgment matrix, the hierarchical single ranking and its consistency check, and the third step is to **determine the weight of the index.**

The target layer is the selection of important indicators for the deployment of UAV take-off and landing points, and the criterion layer is the indicators investigated above. Under the index system, the judgment matrix with 1/3/5/7/9 as the scale is established, the weight of each index is calculated, and the consistency test is carried out to obtain the results.

Construction of a suitability assessment scheme

In the model construction method of road walkability evaluation, Eric T.H. Chan and other experts introduced the application of different decision rules to deduce a variety of walkability scenarios and their potential applicability to different pedestrian groups. Part of the formula is as follows:

WS1 = LAND + SAFE + AEST + AMEN +SUBJ + PATH

WS2 = LAND x SAFE x AEST x AMEN x SUBJ x PATH

WS3 = LAND x SAFE x (AEST + AMEN +SUBJ + PATH)

We will construct the **calculation method of the suitability of UAV take-off and landing point deployment** based on the existing indicators. The suitability index will be obtained by multiplying or adding the index values, the index will be given a coefficient related to the weight, and whether there is mutual compensation between each index will be measured through the weight and the actual application, and different calculation modes will be constructed for different application scenarios.

3.2 Index Weight Calculation

Route index of scenic spot

Service point accessibility and **airdrop cabinet visibility** are important indicators and are key factors for serving people with high mobility. **Degree of functional complexity** has the highest weight among the independent individual items.

Route index of office area

The weight of the **terminal distribution form** index is the largest, which is related to the current situation of the existing traditional distribution mode of home delivery. **Potential user base** and degree of **functional complexity** are important indicators. Office employees can be a stable source of customers, and a variety of functions provide a variety of use scenarios.

Route index of residential area

The weight of the **terminal distribution form** index is the largest, and the weight is higher than that of the same level index in the office area. **Potential user base** and degree of **functional complexity** are important indicators, and the weight of each sub index is equivalent, indicating that more attention should be paid to life services in residential areas.

Dimensions	Criteria	Scenic spot				Office area				Residential area			
		Local Weight	Global Weight			Local Weight	Global Weight			Local Weight	Global Weight		
			Weight	Criteria	Weights		Weight	Criteria	Weights		Weight	Criteria	Weights
a Location and Land use	a1 Type of land use	0.06	0.25	a1	0.02	0.06	0.22	a1	0.01	0.05	0.23	a1	0.01
	a2 District function(within 300m)		0.75	a2	0.05		0.78	a2	0.05		0.77	a2	0.04
b Potential user base	b1 Number of potential users(within 300m)	0.12	0.58	b1	0.07	0.29	0.30	b1	0.09	0.20	0.24	b1	0.05
	b2 Crowd size heat map		0.13	b2	0.02		0.14	b2	0.04		0.23	b2	0.05
	b3 Building density		0.09	b3	0.01		0.14	b3	0.04		0.13	b3	0.03
	b4 Plot ratio near landing points		0.20	b4	0.02		0.43	b4	0.12		0.29	b4	0.06
c Service point accessibility	c1 Number of shops nearby(10-50m)	0.31	0.23	c1	0.07	0.13	0.24	c1	0.03	0.11	0.14	c1	0.02
	c2 Number of shops nearby(within 300m)		0.17	c2	0.05		0.16	c2	0.02		0.11	c2	0.01
	c3 Accessibility to surrounding shops		0.40	c3	0.13		0.37	c3	0.05		0.29	c3	0.03
	c4 Accessibility between restaurants and landing points		0.20	c4	0.06		0.23	c4	0.03		0.46	c4	0.05
d Visibility	d1 Spatial location	0.29	0.21	d1	0.06	0.13	0.16	d1	0.02	0.16	0.21	d1	0.03
	d2 Closest distance to the surrounding building		0.11	d2	0.03		0.13	d2	0.02		0.17	d2	0.03
	d3 Closest distance to the entrance of surrounding building		0.24	d3	0.07		0.38	d3	0.05		0.34	d3	0.05
	d4 Closest distance to the surrounding leisure seat		0.44	d4	0.13		0.33	d4	0.04		0.28	d4	0.04
e Functional complexity	e1 Degree of functional complexity	0.16	0.16	e1	0.16	0.08	0.08	e1	0.08	0.05	0.05	e1	0.05
f Terminal distribution	f1 Terminal distribution form	0.06	0.06	f1	0.06	0.31	0.31	f1	0.31	0.43	0.43	f1	0.43

3.3 Index Value Standard

The unquantifiable indicators such as location and land use and spatial location were eliminated. Combined with the summary of the index values of the existing routes and the influence trend of each index characteristic on the suitability, the index value method in the below table is constructed. Some range values are based on the existing air route spatial data as the reference standard, and other existing standards such as walking life circle and service point radiation range.

	1	2	3	4	5
c1	15 and above	10~14	5~9	1~4	0
c2	56 and above	41~55	26~40	11~25	0~10
c3/m	0~50	51~200	201~350	351~500	500 and above
c4	6 and above	0~5	-5~-1	-10~-6	-11 and below
d2/m	55 and above	36~54	21~35	4~20	0~3
d3/m	80 and above	56~79	31~55	7~30	0~6
d4/m	50 and above	36~49	21~35	6~20	0~5
e1	-2	-1	0	1	2

For **c4 accessibility between restaurants and landing points** index, it is necessary to make a secondary value, and the **walking time, riding time and road obstacle** conditions are used as the value basis.

Value	4	2	0	-2	-4	-6
Riding time	0-2	2-4	4-6	6-8	8-10	>10

Value	4	2	0	-2	-4	-6
Walking time	0-5	5-10	10-15	15-20	20-30	>30

Value	0	-1	-2
Obstacle	Sidewalk	Large scale intersection	overpass

3.4 Suitability of landing point selection

Index processing

As the current operating routes are shown in different scenarios, the a index is not included in the index calculation system. At the same time, the b index is related to the protection of the interests of businesses and the public, and is also an important basis for the layout of air routes at the macro and medium level. Therefore, the a location and land use index and the b potential user base index are **set as the prerequisite screening conditions before the calculation of land use suitability.**

Since the f Terminal distribution index is not the spatial attribute of the land itself, it is an **additional important measure** before and after the placement of the landing point, so the f Terminal distribution index is not included in the spatial calculation model, but **it is discussed in two cases: with and without terminal distribution.**

By determining the overall and local weight of each index, and measuring **whether there is mutual compensation** between each index, the corresponding technical processing of addition or multiplication is carried out. The coefficient of each index is related to the relative value of the weight.

Formula for calculating the suitability of landing points

Scenic spot

$$X_{SP} = c \times d \times e = c_3 \times (c_1 + c_1 + c_4) \times d_4 \times (d_2 + d_3) \times e$$

Office area

HAVE terminal distribution

$$X_{OA1} = (c + d) \times e = (c_1 + c_2 + c_3 + c_4 + d_2 + d_3 + d_4) \times e$$

NO terminal distribution

$$X_{OA2} = c \times d \times e = c_3 \times (c_1 + c_2 + c_4) \times d_3 \times (d_2 + d_4) \times e$$

Residential area

HAVE terminal distribution

$$X_{RA1} = (c + d) \times e = (c_1 + c_2 + c_3 + c_4 + d_2 + d_3 + d_4) \times e$$

NO terminal distribution

$$X_{RA2} = c \times d \times e = c_4 \times (c_1 + c_2 + c_3) \times d_3 \times (d_2 + d_4) \times e$$

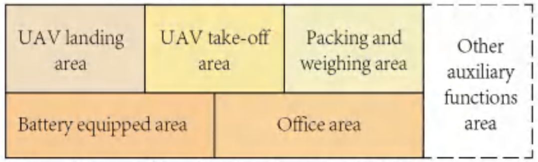
3.5 Suggestions of take-off point selection

Suggested construction index and scope

Dimensions	First grade index	Second grade index	Detailed description
Function layout	Function definition		As an important supplement to the transportation capacity in the region, UAVs cooperate with delivery workers to complete daily delivery work.
	Construction Requirement		The UAV take-off point is recommended to adopt the form of building auxiliary construction, which is recommended to be set at the top floor of the building or the open platform. If you want to set in a crowded place, it is recommended to set it with multi-function services.
			Site coordination: The take-off point should be set close to the large commercial mall.
Supporting facility	Spatial arrangement	functional zoning	Regular take-off points should include: packing and weighing area, UAV take-off area, UAV landing area, battery equipped area, office area and other auxiliary functions. (Refer to the diagram on the right)
Scale of construction	Standard cell fitting scale	Height requirement	It is recommended to set on the roof or open platform of the building, and ensure that there is no shelter in the vertical height within a .radius of at least 3m, centered at the take-off point
		Minimum land area scale	If the integrated take-off and landing airdrop cabinet is equipped, the land area shall not be less than 2mx2m. If a conventional take-off point is configured, the total land use shall not be less than 40m².
		logistics support	It is suggested that the battery equipped area should be set in combination with the UAV take-off area and landing area, while ensuring the safety of electricity consumption and the safety of battery storage at night.
	Location of take-off point	Number of merchants	It is suggested that locations with more than 40 businesses within a radius of 100m should be selected for the deployment of take-off points, and large commercial malls should be preferred for setting.
		Takeaway base order quantity	If the number of businesses within 100m radius is less than 40, but the overall daily takeaway basic order volume is more than 20,000 orders or more places or commercial complexes can also set the departure point.
		Surrounding land use	It is recommended to choose the place with at least 4 types of land use around the location of the departure point.
		Surroundings accessibility	It is preferred to choose large areas with more traffic intersections, pedestrian overpasses or pedestrian underpasses around for the deployment of take-off points.

Layout proposal

Functional partition diagram



Note - Other auxiliary functional areas include but are not limited to setting: leisure seats, educational publicity boards, etc.

In addition to the space requirements on the vertical, the take-off point has less space requirements on the plane. Now the conventional take-off point needs human participation, there are packing and weighing area, office area and other areas. In the future, if it is simplified to **configure the integrated takeoff and landing airdrop cabinet**, only the occupied area of the airdrop cabinet is **2m*2m** and the battery equipped area is required.

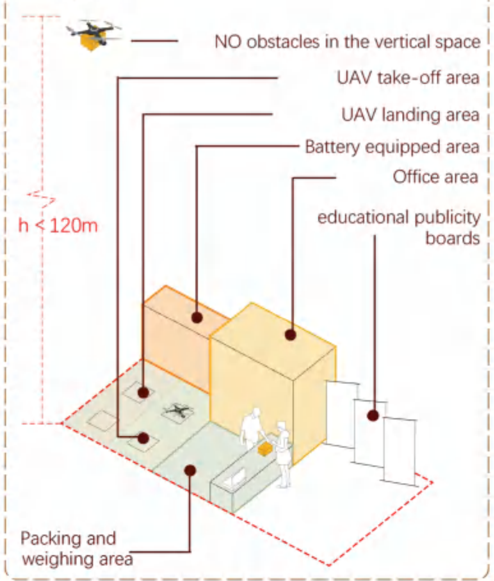
3.6 Suggestions of landing point selection

Suggested construction index and scope

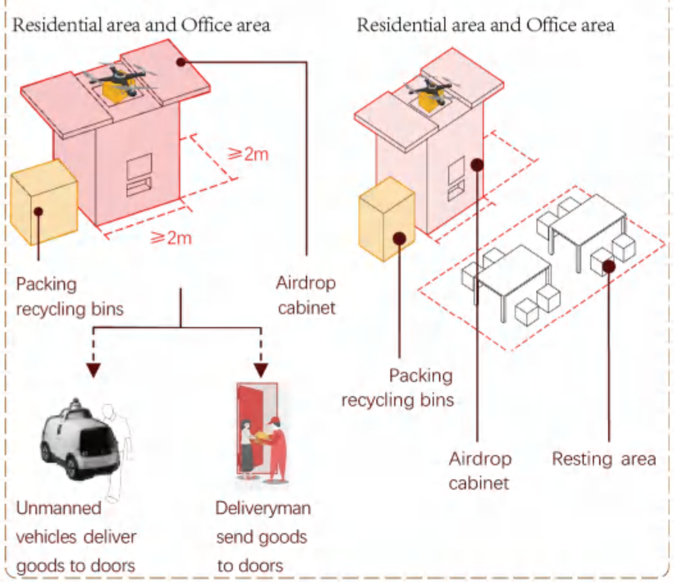
Dimensions	First grade index	Second grade index	Detailed description: Residential area	Detailed description: Office area	Detailed description: Scenic spot
Function layout	Function definition		Provide a landing platform for the UAV and store food for a short time		
	Construction Requirement	Balance of service	The setting of airdrop cabinet should be coordinated with the use of land. The standard unit with the leading function of living, commercial business, industrial development, comprehensive service and scientific and innovative development should be equipped with at least one airdrop cabinet in principle.		
		Site coordination	In the case of no terminal distribution, the airdrop cabinet should have better visibility and accessibility, such as: near the entrance and exit of the community or building, and the traffic station. If there is terminal distribution, it is only required that the location of the airdrop cabinet does not interfere with normal traffic activities.		
Support-ing facility	Spatial arrangement	functional zoning	It is suggested that leisure space such as tables and chairs should be set up around the airdrop cabinet.		
		Height requirement	It is recommended to set on the roof or open platform of the building, and ensure that there is no shelter in the vertical height within a radius of 3m, centered at the take-off point.		
Scale of construction	Standard cell fitting scale	Minimum land area scale	The area occupied by the airdrop cabinet shall not be less than 2mx2m.		
	Location of landing point	Land use and district function	Mainly used for residential function	Mainly used for business and office function	\
		Building density and plot ratio	It is recommended to choose places where building density is of 24% to 57%, and floor area ratio is of 5.4 to 9.6	It is recommended to choose places where building density is of 19% to 52%, and floor area ratio is of 0.5 to 9.4.	\
		Spatial location	It is recommended to set it at the entrance and exit of the community or residential entrance where the flow of people is dense.	It is recommended to set it in the entrance and exit of the building or the entrance and exit of the traffic station around the office area, where the flow of people is dense.	It is recommended to be set in open places or places with landmark attractions, or places with dense crowds.
		Land use complexity	Ensure that there are 2 or more types of land use around the residential area.	It is suggested that there are 2 or more types of land use around the office area.	\
	terminal distribution		According to the actual situation, unmanned vehicles or deliveryman can be configured to deliver food to the door		not commend

Concept image

Take-off point



Landing point



Appendix

1 Questionnaire

Dear Sir/Madam, Hello!

We are students from the School of Architecture and Urban Planning of Shenzhen University, and a research team on the market demand of UAV takeout delivery service. This survey is mainly to understand your cognition and usage demand of UAV distribution. The results of the survey will help promote the reform of the logistics industry and the development of unmanned distribution. I hope to get your support and assistance.

Introduction to UAV takeout delivery: that is, by operating an unmanned aerial vehicle, the packed takeout will be automatically delivered to the takeout cabinet at the landing point from the starting delivery point. At present, UAV has the advantages of long distance, short time (3 km and 15 minutes), high safety, no traffic congestion on the ground, and no delivery time limit (24h). At the same time, it also has some shortcomings such as large noise and small single distribution capacity

1.Your gender is?
A. Male B.Female

2.What is your age?
A. 18 and below B. 18-25 C. 26-35 D. 36-50 E. 50 and above

3.Which of the following are you?
A. Live nearby B. Work nearby
C. I often pass by here D. Occasionally pass by or play

4.How many times a week do you order takeout on average?
A. 10 and above B. 8-10 C. 5-7 D. 3-5 E. 3 and below

5.When you often use instant delivery (such as takeout, Internet supermarket, etc.)?
A. Breakfast time B. Lunch time C. Dinner time D. Others (morning, afternoon. evening,etc.) ________

6. Have you used the UAV delivery/delivery service here?
A. Never B. 3 times and below C. 3-5 times E. 5-7times F. 7 and above

7. The UAV delivery time of your order is usually?
A. Never used B. 15 min and below C. 15-30min D. 30-45min E. 46-60min F. 60 min and above

8. Which of the following items would you most like to be delivered by UAVs?
A.Meals B.Desert and drinks C.Daily used items D.Vegetables and fruits E.Medicine G. Others________

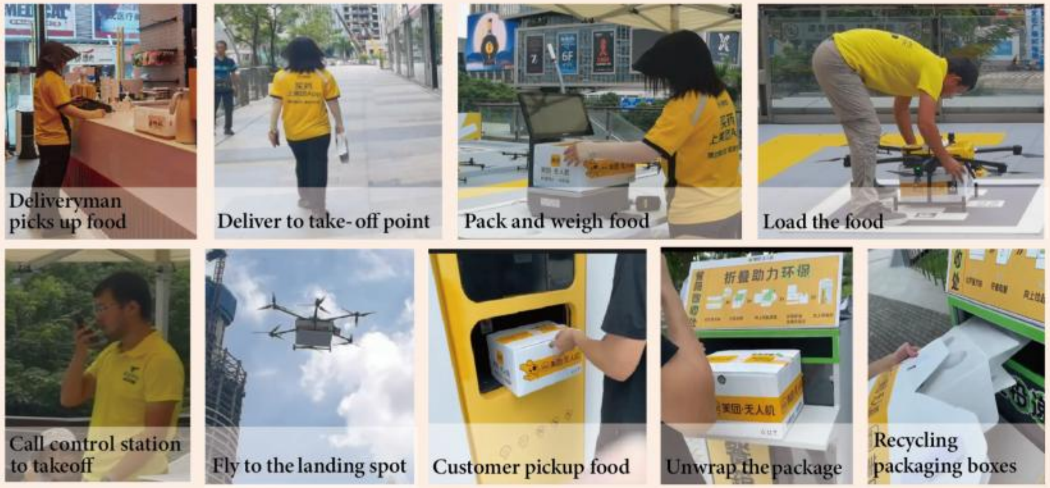
9. What is your opinion on UAV food delivery/express delivery service?
A. Very good B. Good C. Normal D.Bad E.Very bad

10. What are reasons make you might want to accept drone delivery?
A. Short and guaranteed delivery time (free from traffic jams) B. Long distance delivery
C.Reduce lost takeout D. Low breakage rate
E. 24hrs delivery F. No need to deal with people G. Others________

11.What are reasons make you might not want to accept drone delivery?
A.Trouble placing or picking up process B.It can't be delivered home
C.Delivery fees are high D.The selection is limited
E.Threat to pedestrians or property on the ground F.Worry about line of sight, noise interference
G.Fear of disclosure of personal privacy and purchase behavior H. Others________

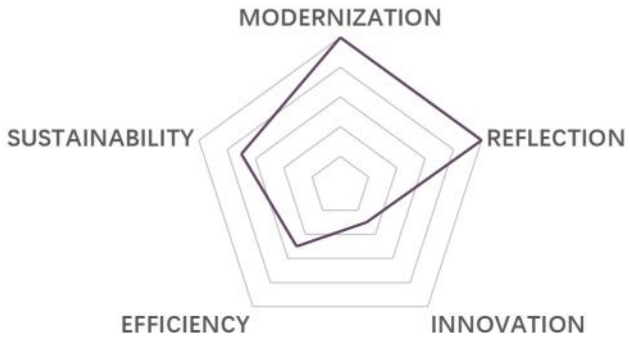
12.If there are more UAV stations in the future, where would you like to deploy them?
A.Downstairs in the residential area B.Downstairs of the office building
C.Hotel downstairs D.Parks E.Tourist attractions F. Traffic stations G.Express station H.Others________

2 Whole process of UAV delivery



02 TOP DOWN AND BOTTOM UP

Two Perspectives to Explore Urban Structure



Type: Academic work | *Principles of urban and rural planning (3)*
Instructor: Wuyang Hong
Date: 09/2023-01/2024
Location: Shenzhen, China
Individual work

Urban planning needs reflection. Cities are the result of various forces, such as self-development and planning constraints, so urban planning cannot be merely on paper. Cities need regular "physical examination" to detect urban problems and analyze the gap between planning and reality, so as to lay the foundation for formulating more reasonable and effective urban planning.

In the past era of speedy urban development, in order to rapidly improve urban functions and stimulate market vitality, China's urban development mostly adopted the top-down model led by the government, which also caused problems such as blocking the urban context and greater social resistance to project promotion. In the current practice of urban renewal in China, there is a trend from top-down to a combination of bottom-up and top-down.

Aiming at the problem of **urban center structure**, this study uses **POI data** and **k-means clustering analysis** to explore the big difference between Shenzhen's planning and the actual development of the city. At the same time, the reasons for the differences are analyzed, and the policy suggestions are given.

1 Introduction

1.1 Background

The opinion that both top-down and bottom-up forces can influence urban development are mentioned in many urban studies, such as Castagnoli's "irregular city" and "uniform scale city", Pierre Lavedan's "Created city" and "random city", Kostof's "planned city" and "unplanned city".[1] Basically, "top-down" is a kind of authoritative governance[2], which is led by the government or jointly led by the government and market entities, and takes blueprint-oriented urban renewal mechanism as the goal. [3]By contrast, "bottom-up" often descibes new, apparently spontaneous development initiatives pursued by market-enabled town and municipal governments independent of the existing "top-down" socialist planning apparatus,[4]which can reflect the needs of the public better.

In the past era of speedy urban development, in order to rapidly improve urban functions and stimulate market vitality, China's urban development mostly adopted the top-down model led by the government, which also caused problems such as blocking the urban context and greater social resistance to project promotion.
In the current practice of urban renewal in China, there is a trend from top-down to a combination of bottom-up and top-down.

1.2 Research gap and Purpose

At present, some scholars have confirmed that in the dimensions of scale, territory, networks and temporality, top-down and bottom-up, planned and unplanned, are mutually constituted in all instances of settlement transition.[5]However, few studies have explored the extent to which urban spaces have evolved in accordance with government blueprints and goals in the context of China's top-down model. For China and many other countries with strong governments, assessing planning goals and the status quo of urban development remains an important issue.
The combination of top-down and bottom-up is an innovative urban development model, which emphasizes the cooperation of multiple subjects. Some scholars have confirmed that the settlements that have been formed in spontaneous have some derived properties which can enrich the pool of possible solutions traditionally achieved with top-down planning.[6]However, since such feedback should be tailored to local conditions, there are not enough studies in China to systematically provide bottom-up planning feedback and suggestions for top-down planning.

1.3 Study area

Shenzhen is a representative city of the reform and opening up in China, as well as a pioneering demonstration zone of socialism with Chinese characteristics. Shenzhen has developed rapidly since the 1980s with policy support, and it even surpassed Guangzhou and Hong Kong and became the city with the third highest GDP in China in 2019. However, compared with other big cities, Shenzhen only covered a small area. Its land area is only 1997 square kilometers, which is even less than one eighth of the area of the capital Beijing. One of the important reasons that Shenzhen rapidly develop into an international metropolis in 40 years is the strong policies and planning blueprints. It can be said that Shenzhen is one of the regions that best embodies "top-down" urban planning. For Shenzhen, studying the degree and pattern of top-down's impact on the city is one of the most important topics for the government.

2 Materials

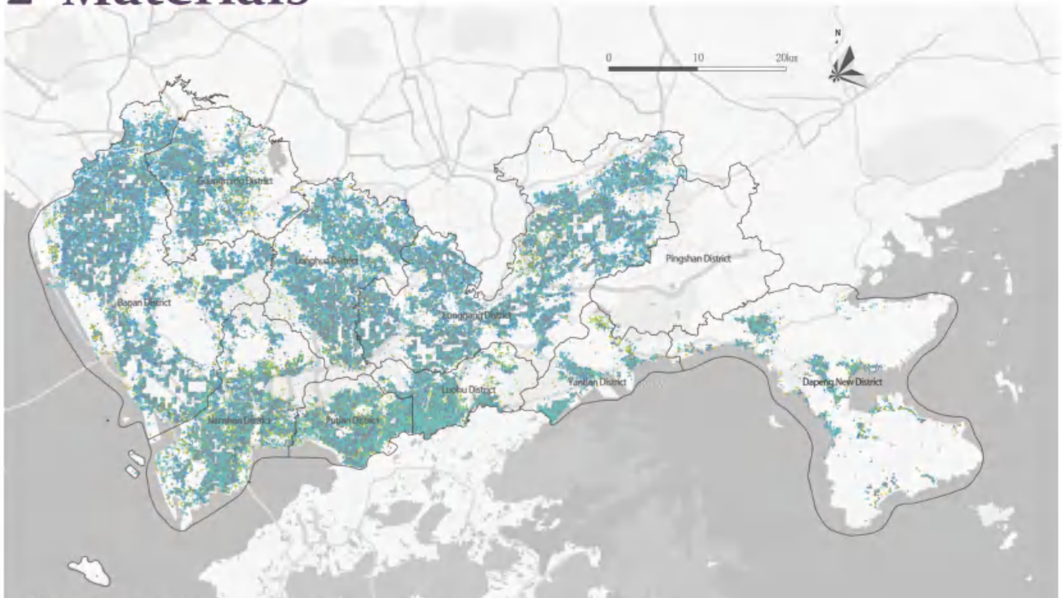


Fig1. Overview of the data and study area

2.1 Data preparation and Software

Shenzhen is a mega-city with many functions. In order to examine the implementation of top-down forces, the following data needs to be prepared: the basic grid map, and POI data which can represent functions of civic centers, such as urban services and businesses.
The raw POl data are obtained from the Baidu Map website using Python crawlers., properties of which include name, latitude and longitude coordinates,fctional category ofthe POI, etc.The study use QGiS (ver. 3.38.0)to spatially visualize the various data, and use EPSG:4326-WGS 84 as the unifed coordinate system.

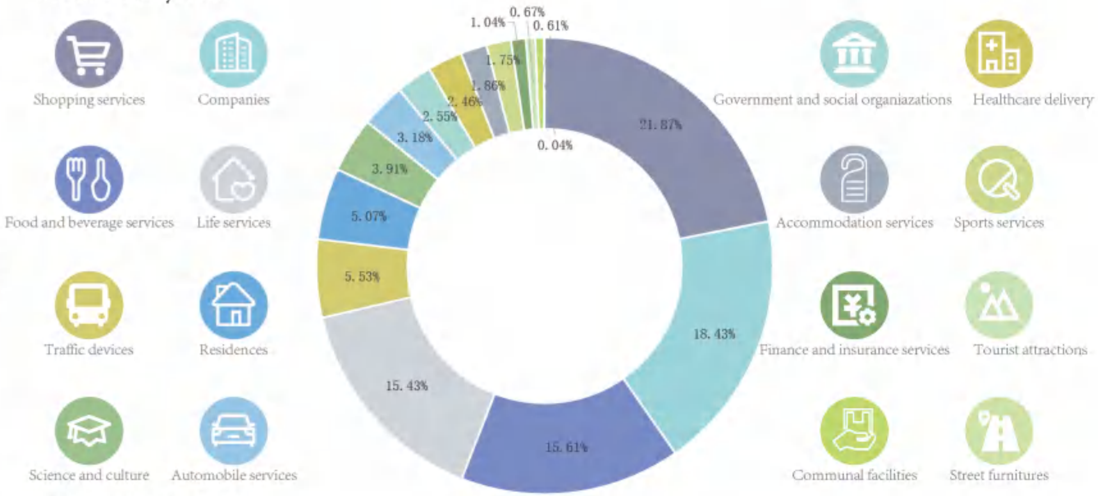


Fig2. Data composition

Data selection

Territorial Spatial Master Planning of Shenzhen(2020-2035) describes the urban center system as a "multi-center spatial pattern with functional differentiation of labor and cooperation" when planning the urban function center. Combined with the description of the functions in the the district-level urban development pattern planning maps, the research divides the urban centers into the following categories:

Fig3. Extraction of city center function

The next task is to select the specific POI data representing the eight kinds of functions of city centers.

FUNCTIONS OF CITY CENTERS	SELECTIONS OF POI		
	BASETYPE	SUBTYPE	OTHER FILTERS
Culture and Education	Companies	filtrating companies in culture industry	
	Science and culture		
Commercial retail	Food and beverage services		
	Shopping services		
	Life services	delete "travel agency" "bath massage place"	
Finance	Finance and insurance services		
Foreign trade	Companies	filtrating transnational corporation	
Leisure tourism	Food and beverage services		
	Tourist attractions		
	Life services	select "travel agency" "bath massage place"	
	Accommodation services		
Administration and Management	Government and social organiazations		
Production	Companies	select "factory"	
Technovation	Companies	filtrating companies in science and technology industry	
	Science and culture	select "scientific research institution" "school"	

Table1. Data processing flow

2.2 Methods

Clustering is a key unsupervised learning method used to classify similar data points and explore the intrinsic structure of the data By revealing dense and sparse regions in a dataset, cluster analysis helps researchers identify global distribution patterns and complex data attribute associations.[7]

K-means is one of the most widely used clustering algorithms, suitable for processing large-scale data sets. The method relocates cluster centers by iteratively calculating the average position of cluster centers. The classical algorithm flow consists of the following steps:

Initialize → Distribute data points → Update cluster centers → Iterative optimize

(1)Initialize

Randomly select k data points as initial clustering centers. In this study, the **Elbow Method** is employed to determine the value of k. For each value of k ranging from 1 to 15, the k-means clustering algorithm is performed, and the **Sum of Squared Errors (SSE)** is calculated separately.

Plot the curve with k values on the horizontal axis and the corresponding SSE values on the vertical axis. As shown in the fig4, in this study, three inflection points, commonly referred to as "elbows", appear at k=2, k=4, and k=6, which indicates that at these k values, the clustering results have good cohesion without being overly complex.
Considering the diverse functional compositions of urban centers in this study, the maximum value among these three, k=6, is chosen.

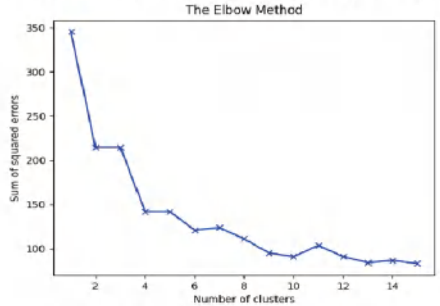


Fig4. Elbow method result

(2)Distribute data points

For each data point, calculate its distance to each clustering center and assign it to the cluster of the nearest clustering center.

(3)Update cluster centers

After the data points have been assigned, the next step is to update the center of each cluster. Calculate the mean of each cluster and use it as the new clustering center.

(4)Iterative optimize

Repeat step 2 and 3 until the positions of the clustering centers stabilize or the predetermined number of iterations is reached, which indicates that the algorithm has converged.
In short, this study selects the number of clusters as 6, and use "Attribute based clustering" plugin (2.2.1 version) to implement the above experimental steps.

3 Results and Discussions

3.1 Experiment Results

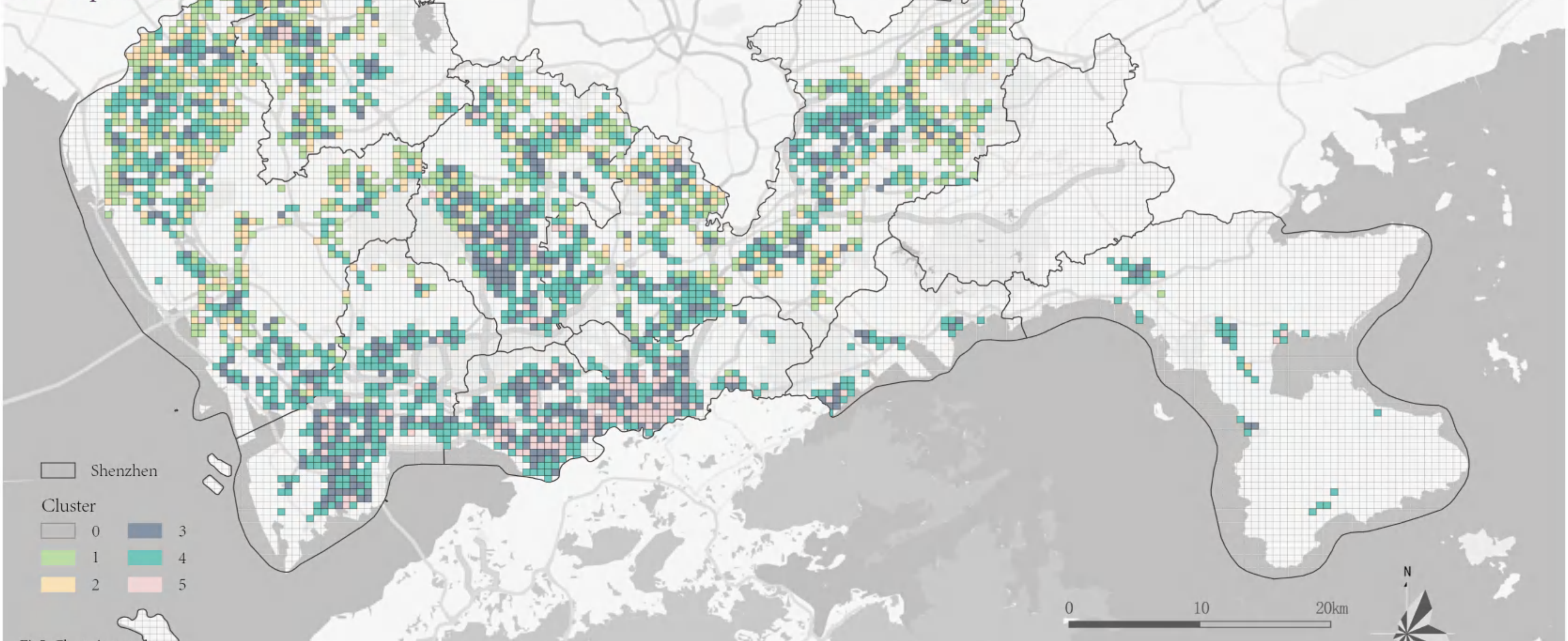


Fig5. Clustering results

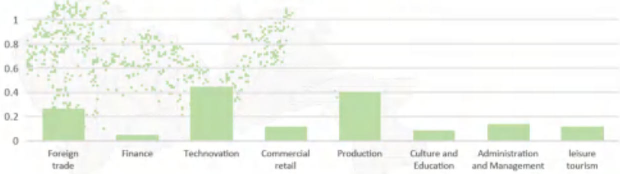
For the aggregation of each function in the grid, the research does normalization processing on the basis of average value, and the formula is as follows:

$X^* = (X - min) / (max - min)$

Where $\boldsymbol{X^*}$ represents the normalized value of the aggregation of a certain function within a cluster; $\boldsymbol{X}$ denotes the original average value of the aggregation, $\boldsymbol{min}$ signifies the minimum value of the aggregation for that function across all clusters, and max denotes the maximum value of the aggregation degree for that function across all clusters.

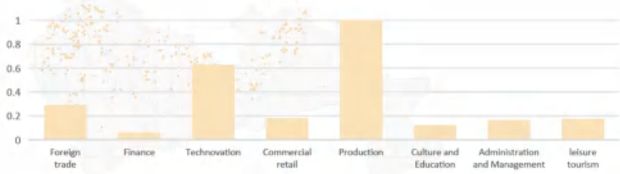
Cluster 1

total aggregation index=1.63



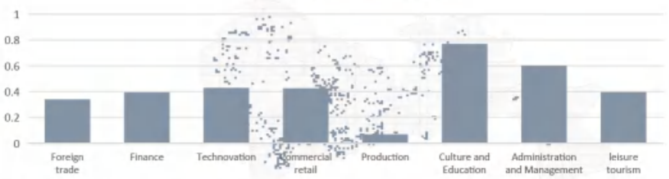
Cluster 2

total aggregation index=2.62



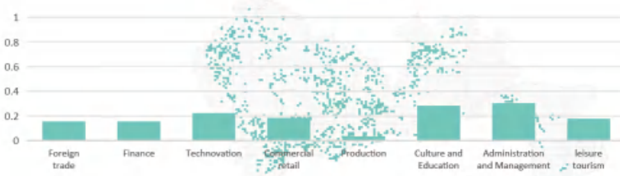
Cluster 3

total aggregation index=3.43



Cluster 4

total aggregation index=1.51



Cluster 5

total aggregation index=7.06

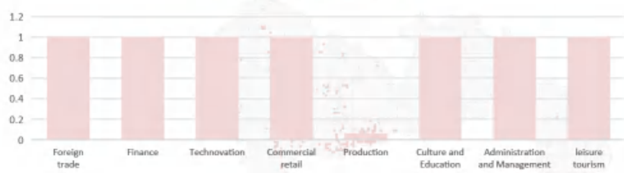
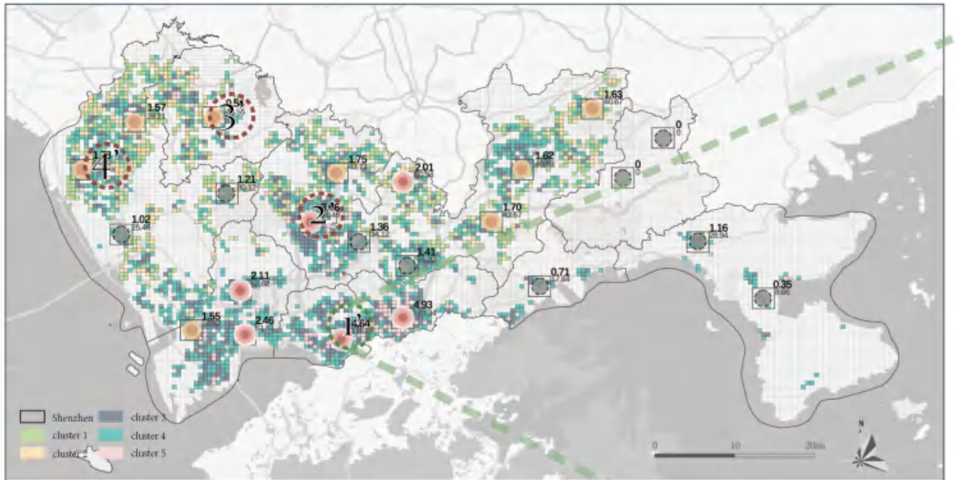


Fig5. Meaning of clusters

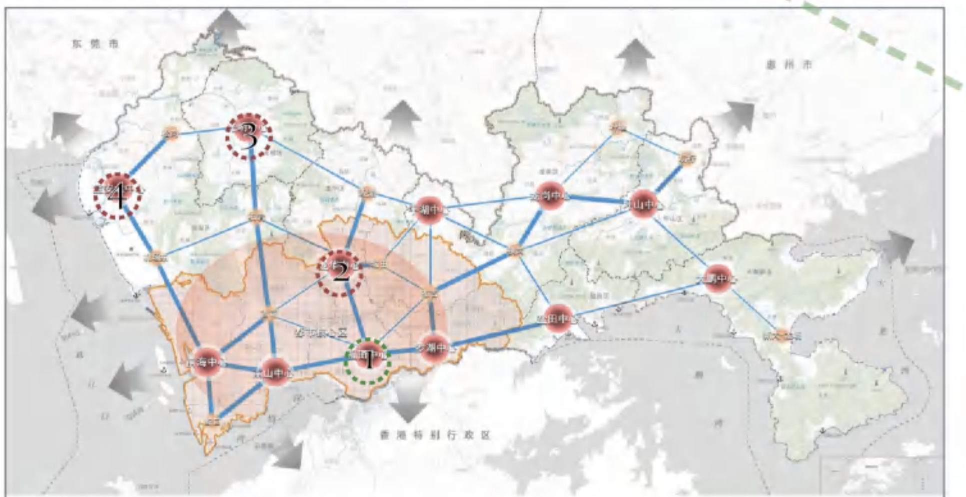
3.2 Discussions

Grading city centers



Node with the best foundation
Node with good foundation
Node with bad foundation
3.03 Average aggregation index of one node
75.71 Total aggregation index of one node
Scope of one node

Fig6.*Bottom up:* Result from clustering



Shenzhen
Urban core area
Axis of function connection
Urban function center
Urban function node

Fig7.***Top down:*** Central system planning
from Territorial Spatial Master Planning of Shenzhen(2020-2035)

The research defines the influence range of a center(node) as **5*5 grids**, that which means 2500 m *2500 m. The average aggregation of all grids within the range is taken as the aggregation index of the center(node), so as to evaluate the development status and potential of the center(node).

The results show that the centers(nodes) with aggregation index **less than 1.5** have poor development conditions, so they can be excluded. Centers(nodes) with aggregation index **values between 1.5 and 2** have good development conditions and can be considered as urban function nodes. The centers(nodes) whose aggregation index are **above 2** have the best development foundations and have the potential to develop into a municipal functional center.

Classifying city centers

1/1' Futian center

Statement from plannings

"WORLD FINANCIAL CITY"
"INTERNATIONAL FASHION CITY"
"SCIENCE AND TECHNOLOGY INNOVATION ENGINE"
"ADMINISTRATIVE AND CULTURE CENTER"
"High-end business"
"International contacts"
"Public culture"
"Consume district"

function of city center

- Culture and Education
- Commercial retail
- Finance
- Foreign trade
- Technovation
- Administration and Management

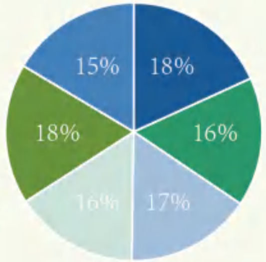


Fig8. Function ratio of Futian center

2/2' Longhua center

Statement from plannings

"Core area of digital economy"
"High-tech industries, advanced manufacturing as well as cultural and creative industries gathering area"
"Characteristic, digital fashion consumption vitality area"

function of city center

- Culture and Education
- Commercial retail
- Production
- Technovation+Finance
- Technovation

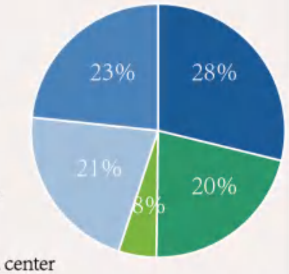


Fig9. Function ratio of Longhua center

In the function ratio of Longhua center, the production is in a weak position compared with other functions, so the leading responsibility of this function in the center should be removed.

3/3' Guangming center

Statement from plannings

"Comprehensive national science center core area"
"Set headquarters office, science and technology services, cultural exchanges, ecological leisure, transportation hub in one"

function of city center

- Culture and Education
- Technovation
- Finance+Foreign trade
- Leisure tourism

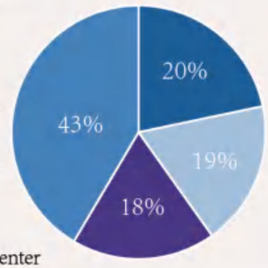


Fig10. Function ratio of Futian center

The culture and education function of the Guangming center is obvious, so it should be considered as the leading function of the center, and other industries and functions can be developed on the basis of it.

4/4' Bao'anbei center

Statement from plannings

"Intelligent manufacturing, international convention and exhibition industry, airport economy and Marine industry cluster area"
"Advanced manufacturing park"
"Emerging industry cluster"
"Innovation headquarters area"

function of city center

- Culture and Education
- Technovation
- Finance
- Production
- Commercial retail+Foreign trade

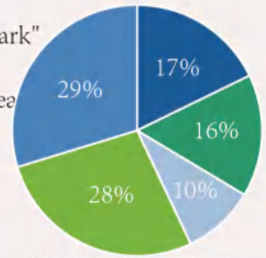


Fig11. Function ratio of Bao'anbei center

The financial function of Bao'anbei center is difficult to occupy a dominant position, so it should be considered to build a regional function dominated by technology and innovation as well as production.

Reasoning the gap between Top down and Bottom up

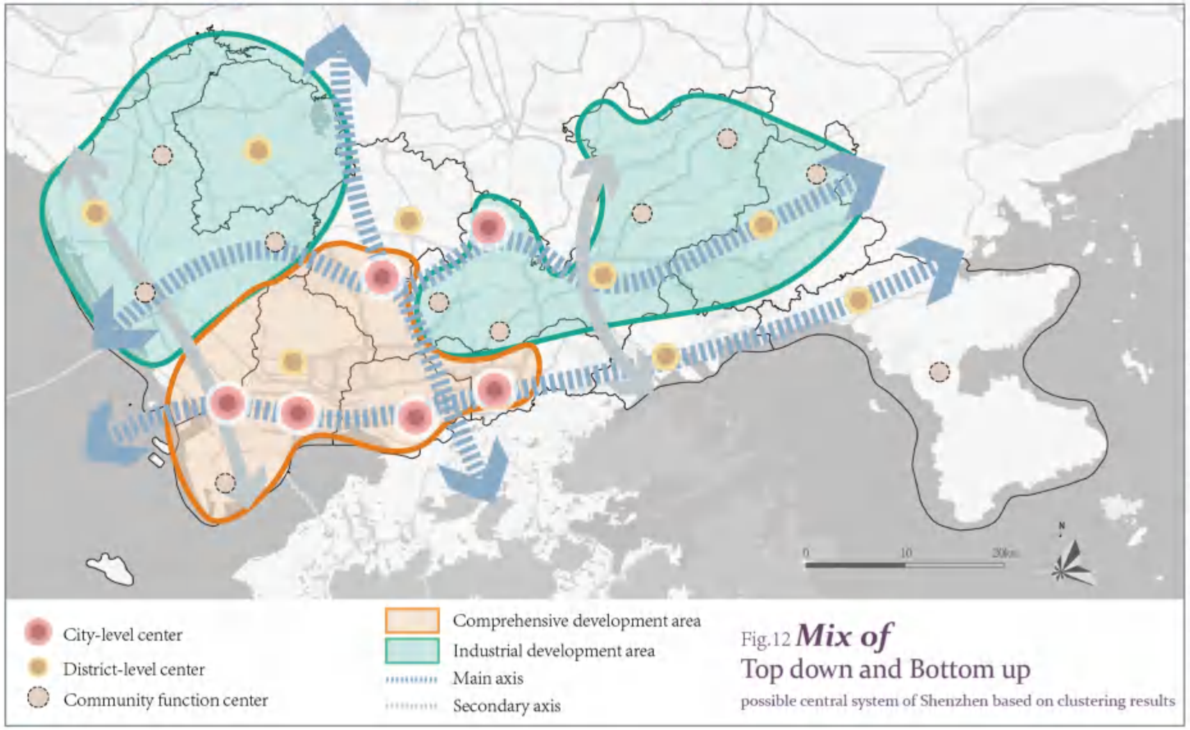
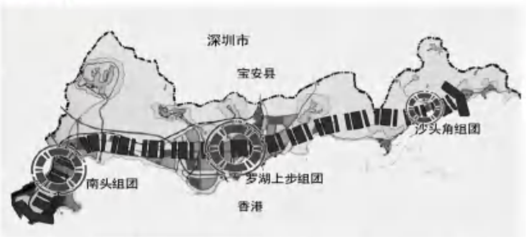
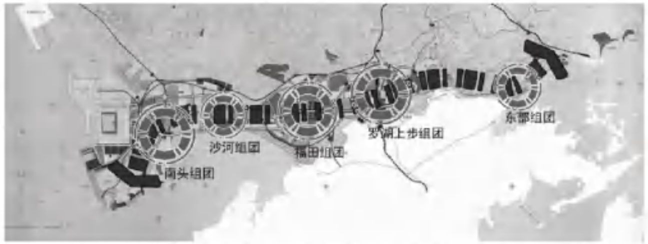


Fig.12 **Mix of Top down and Bottom up**
possible central system of Shenzhen based on clustering results

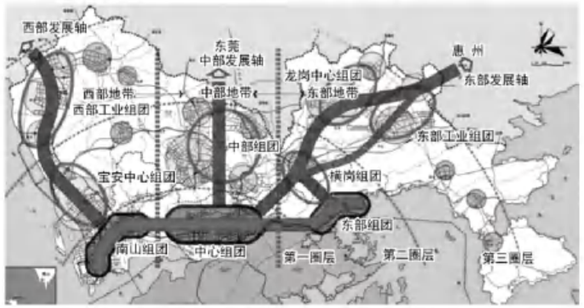
Policy



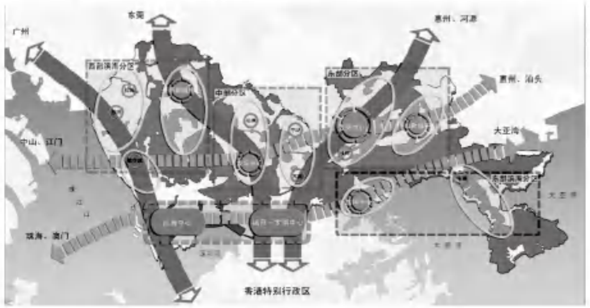
Master Plan for Shenzhen Special Economic Zone, 1982



Master Plan for Shenzhen Special Economic Zone, 1986—2000



The Comprehensive Plan of Shenzhen City, 1996—2010



The Comprehensive Plan of Shenzhen City, 2010—2020

Fig13. Overall plan of Shenzhen City over the years

Unlike large cities such as Beijing and Shanghai, Shenzhen's urban center system is greatly influenced by policies and urban planning.
As a front of reform and opening up, Shenzhen's development is strongly influenced by policies. Moreover, Shenzhen only began to develop rapidly as a special economic zone in the 1980s, when China's urban planning had already formed a certain system. At that time, All the advanced urban planning theories are concentrated in Shenzhen for testing. It can be said that Shenzhen lacks the opportunity of independent evolution, and its development and change are greatly affected by policies
However, the influence of policy planning factors on employment centers in Beijing and Shanghai is not obvious, and they are greatly influenced by economic structure. Driven by the service industry, the sub-centers of employment in Beijing are mainly distributed near the main centers of the city. In Shanghai, driven by the service and manufacturing industries, the sub-centers of employment are far away from the main centers of the city.
Under the guidance of urban multi-center system planning, Shenzhen has formed a three-level center system of main center, sub-center and group center.

Land Ownership

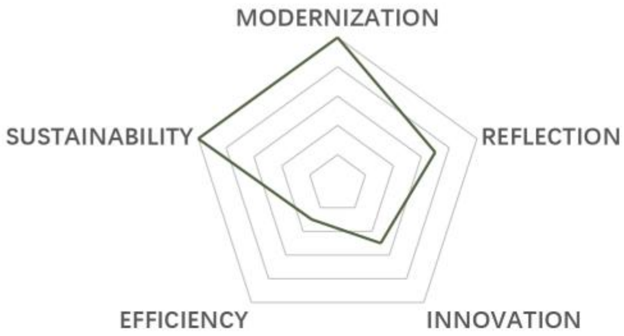
Due to the particularity of the policy, the complicated land policy leads to the existence of a large number of land with unknown ownership in Shenzhen. In 1992 and 2004, Shenzhen carried out the procedures of unified expropriation and unified transfer of collective land, indicating that there are no more rural areas in Shenzhen. However, many former villages have not completely completed this procedure. In areas where land expropriation procedures were not completed in the early stage, due to economic development, the villagers' requirements for land expropriation income increased greatly in the later stage. Besides, there are many phenomena of "planting housing security land", that is, villagers rush to build all kinds of illegal buildings, so that "premises are separated".
In disguise, the floor area ratio of informal housing (urban villages) in Shenzhen has increased rapidly, and many areas have reached more than 4. Therefore, Shenzhen is lack of lands and the city is uneven-developed. Even in the city center, there are many such high-density and scattered urban village housing.

Location and Transportation

The proximity of Futian and Luohu Centres to Hong Kong gives them a natural advantage in economic development. It facilitates the attraction of Hong Kong and international capital and resources, and promotes the development of services such as finance and commerce. There are two important strategic cooperation platforms between Shenzhen and Hong Kong: Qianhai Shenzhen-Hong Kong modern Service Industry Cooperation Zone, located in Qianhai center; One is the Hetao Shenzhen-Hong Kong Science and Technology Innovation Cooperation Zone, located near Futian and Luohu centres. Other centres are difficult to develop due to the lack of cooperation with Hong Kong and distance from the city centre.
Besides, The railway hub's service coverage of the city is insufficient, and there is a trend of "heavy in the west and light in the east". On the one hand, the western part of Shenzhen is over-developed through transportation, while the eastern part is relatively underdeveloped due to the lack of railways. At the same time, the urban rail transit density is also unevenly distributed, especially in the southern and central parts, while the northern and two wings are extremely sparse, and the central system is difficult to develop evenly.

03 GREENWAY IS ENDANGERED

Assessing and Restoring Ecological Corridor Connectivity in Shenzhen Using Spatial Data



Type: Academic work | *Ecological restoration planning of territorial space*
Instructor: Wuyang Hong
Date: 02/2024-06/2024
Location: Shenzhen, China
Individual work

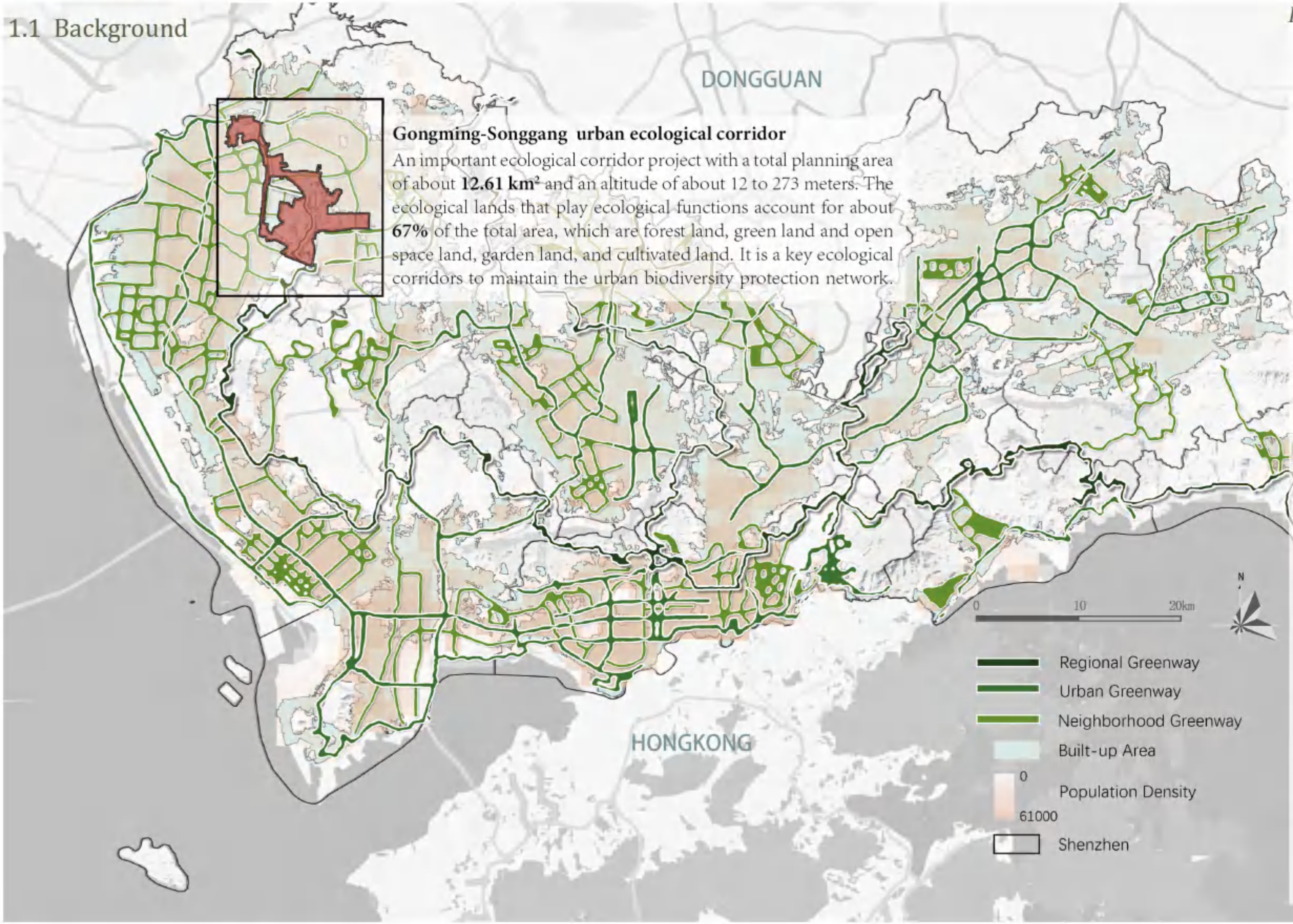
Urban planning needs sustainability. The rapid development of cities often sacrifices the ecological environment, so that the living space of animals and plants will be squeezed and the ecological environment will be destroyed. Therefore, even if the city needs to grow and expand, it is necessary to set aside some green access to maintain the stability of the ecosystem.

Greenway breakpoint problem refers to the fact that in the construction of urban greenway, due to various reasons, the greenway cannot form a continuous linear path, which affects the coherence and use experience of the greenway. In addition, the planning and construction of greenway network in some areas has become a "face project", with problems such as single function and business form, lack of ecology and lagging infrastructure construction. These greenways are only "roads" and cannot play ecological functions.

In this study, **a method to identify the breakpoint** of urban greenway has been proposed and **renovation suggestions** for four typical breakpoint types are given.

1 Introduction

1.1 Background



Gongming-Songgang urban ecological corridor

An important ecological corridor project with a total planning area of about **12.61 km²** and an altitude of about 12 to 273 meters. The ecological lands that play ecological functions account for about **67%** of the total area, which are forest land, green land and open space land, garden land, and cultivated land. It is a key ecological corridors to maintain the urban biodiversity protection network.

Introduction of the greenway system

Regional Greenway

The greenway connects the cities and cities in the region, which can also connect the important natural, cultural and recreational resources of the region.

Urban Greenway

The greenway connects important functional groups in the city, mainly connecting various blue and green open Spaces and important natural and cultural nodes.

Neighborhood Greenway

The greenway that connects various communities and green Spaces within the city to serve surrounding residents

History of greenway construction in Shenzhen

Start up

· In 2010, Shenzhen **started greenway construction**, and proposed that 300km of provincial greenway, 500km of urban greenway and 1200km of neighborhood greenway will be built in the next five years, with a **total length of 2000km**.

· From January 2011 to August 2011, Shenzhen completed the construction task of all regional greenways in the city, and **started the interconnection project** of urban greenways, neighborhood greenways and regional greenways

Interconnection

· In December 2015, regional greenways, urban greenways and neighborhood greenways in the city were **fully connected** . By the end of 2012, the regional greenway has become mature and perfect, and the operation mechanism is sound.

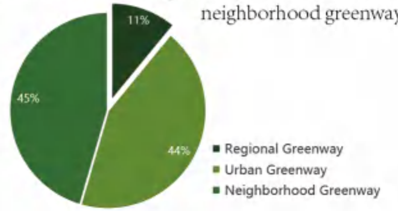
· By the end of 2015, urban greenways and neighborhood greenways have **been basically completed**, forming a **greenway network system**.

Quality Improvement

· In 2019, Shenzhen has built a greenway network with a total length of about **2,448 km**, with a greenway density of **more than 1.2 km/km².** Meanwhile, a new round of **greenway quality improvement** will be launched, and more high-quality greenways will bring more quality slow life to Shenzhen citizens

· It is expected that by 2025, Shenzhen will form **the basic framework of a park city**, and initially establish a comprehensive park city system and a backbone network of trails throughout the city, with more than **1,350 parks** of various types and **more than 4,000 kilometers of greenways.**

By May 2024, Shenzhen has built 3,406 kilometers of greenways , including 373 km of regional greenways, 1,484 km of urban greenways and 1,549 km of neighborhood greenways.



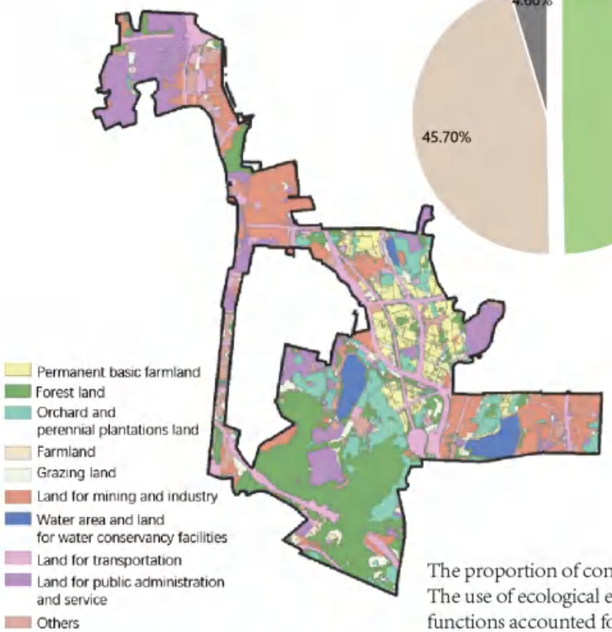
Greenway breakpoint problem refers to the fact that in the construction of urban greenway, due to various reasons, the greenway cannot form a continuous linear path, which affects the coherence and use experience of the greenway. There are various reasons for its formation, mainly for the motor vehicle section barrier, but also engineering facilities barrier, management barrier and imperfect marking system. In addition, the planning and construction of greenway network in some areas has become a "face project", with problems such as single function and business form, lack of ecology and lagging infrastructure construction. These greenways are only "roads" and **cannot play ecological functions**

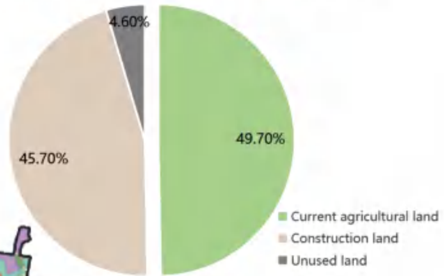
1.2 Study Area

Poor regional ecological connectivity

There are **three breakpoints** in Gongming-Songgang urban ecological corridor, and the gap lengths of ecological elements are 1400 meters, 1618 meters and 1143 meters respectively. The **fracture degree index of the corridor reaches 50%**, and the ecological process between the two nearest ecological cores is not fundamentally connected, and the basic function of the corridor is lost. There are certain challenges in ecological clearance and restoration of the corridor fault section. There are total of **18 roads** with different levels cutting the corridor, which aggravated the ecological isolation and threatened the safety of key species. The main roads were larger than the two-way 4-lane roads, which **blocked the activities of key species** such as ocelot.

Corridor fracture degree = construction land area in corridor/total corridor area =0.46





The proportion of construction land is high. The use of ecological elements with ecological functions accounted for about 67% of the area.

Inefficient industrial parks account for the majority

Urban construction is dominated by low-quality and inefficient industrial parks. At present, most of the industrial parks are constructed by community collective economic organizations, including Shanmen Industrial Zone in Yanluo Street and Lougang International Industrial zone in Songgang Street, which are all **located outside the industrial block line**. There is **only one enterprise above designated size**, and the rest are small processing plants with low technology content, small scale and scattered distribution. A small number of original rural residential buildings are distributed around the Wuzhirao reservoir.

2 Materials

2.1 Methods

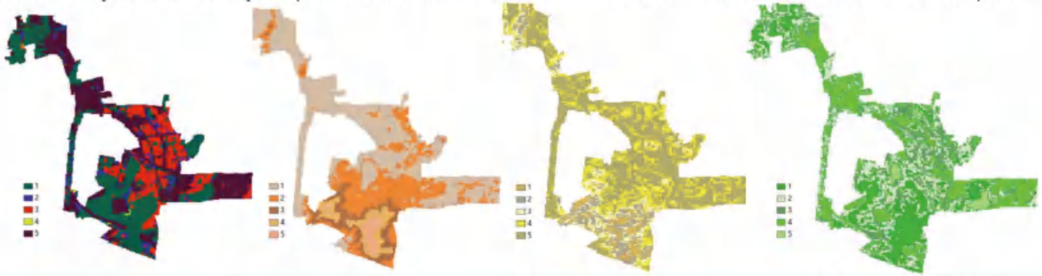
Minimum Cumulative Resistance Model (MCR) is mainly used to describe and predict the movement path of objects or individuals in space. The core idea of the MCR model is that a moving object (e.g., an animal, a vehicle, etc.) will choose a path that minimizes the cumulative drag it experiences during its movement. The "resistance" here can be understood as the difficulties or obstacles faced by the object when moving in different environments, and these resistance factors may include terrain, road conditions, distribution of other objects, etc.
The MCR model was first proposed by KNAAPEN et al. and used in landscape planning. The essence of the MCR model is the process of ecological flow overcoming the resistance of different landscape elements in the process of spatial coverage. It can comprehensively consider the internal relationship of ecological processes and reflect the potential trend of species movement, so as to judge the connectivity of target unit and source unit.
By selecting the **resistance factors**, the **evaluation index system of ecological resistance** was established. At the same time, the existing research is integrated to determine the **weight of each factor** in order to realize the construction of MCR model。

GOAL	CRITERIA	ALTERNATIVES	FORMULA OF FRACTURE DEGREE
EVALUATION INDEX SYSTEM OF ECOLOGICAL RESISTANCE	LAND CONDITIONS	LAND USE TYPE	$E = \sum_{j=1}^{m} p_{ij} w_j$
		NORMALIZED DIFFERENCE VEGETATION INDEX	
	TOPOGRAPHIC CONDITIONS	DIGITAL ELEVATION MODEL	
		SLOPE	

E is the fracture degree index of the ith evaluation unit (grid), and P_{ij} is the jth index of the ith unit. W_j is the weight of each index.

2.2 Data preparation and Software

In order to calculate the **fracture degree** of the ecological corridor and **identify the breakpoint**, the following four kinds of data are needed, which are **land use type(LUT),digital elevation model （DEM）, slope(SL) and normalized difference vegetation index （NDVI）**.
The land use data were from the third national land resources surveys(2020) provided by Bureau of Planning, Land and Resources of Shenzhen Municipality. The DEM data used the 30m resolution GDEMV2 digital elevation data, downloaded rom Geospatial Data Cloud Platform of Computer Network Information Center, Chinese Academy of Sciences (https://www.gscloud.cn). The slope data is based on the above DEM data, and it is analyzed and processed according to the research needs. NDVI data is based on Landsat 8 OLI_TIRS satellite images downloaded from Geospatial Data cloud platform with a resolution of 30m, which were analyzed and processed according to the needs of the study.
The study use ArcMap (ver. 10.8.1)to spatially visualize the various data, and use EPSG:4326-WGS 84 as the unifed coordinate system.



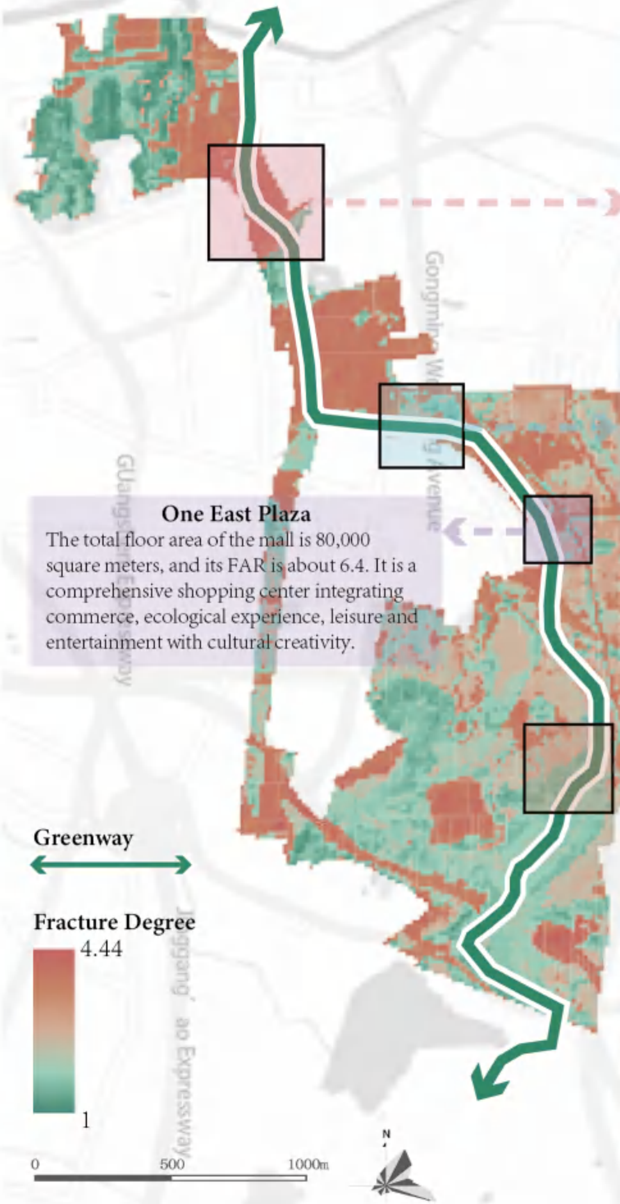
LUT | DEM | SL | NDVI

RESISTANCE VALUE	LUT	DEM	SL	NDVI
1	Waters and Woodlands	[146,225]	[24.7,+∞)	[163,209]
2	Grassland	[96,146)	[16.7,24.7)	[129,163)
3	Farmland, Orchard and perennial plantations	[54,96)	[9.38,16.7)	[96,129)
4	Unused land	[26,54)	[3.84,9.38)	[56,96)
5	Construction land	[4,26)	[0,3.84)	[0,56)
WEIGHT	**0.39**	**0.21**	**0.14**	**0.26**

3 Results and Discussions

The area with the largest fracture degree is mostly the area with the concentration of industrial land, and some areas, such as the Shanmen industrial park and Lougang industrial park, have a large resistance surface area, where the corridor is completely blocked, and the biological passage is extremely difficult.

In this study, four typical land use types that are easy to lead to high fracture degree are selected at the breakpoint, which are industrial park, shopping mall, urban village and residential area. The typical reasons for the fracture of the greenway are analyzed respectively, and a standardized design is made to solve this problem, which provides a reference for the method of greenway restoration.



Shanmen Industrial Park

The industrial park covers an area of about 28,000 square meters with a floor area ratio (FAR)of about 4.5. By the end of 2022, there are 45 enterprises in the park, 22.2% of which are distributed in the computer, communication and other electronic equipment manufacturing industry, and 17.8% in the wholesale industry.

Yulv Village

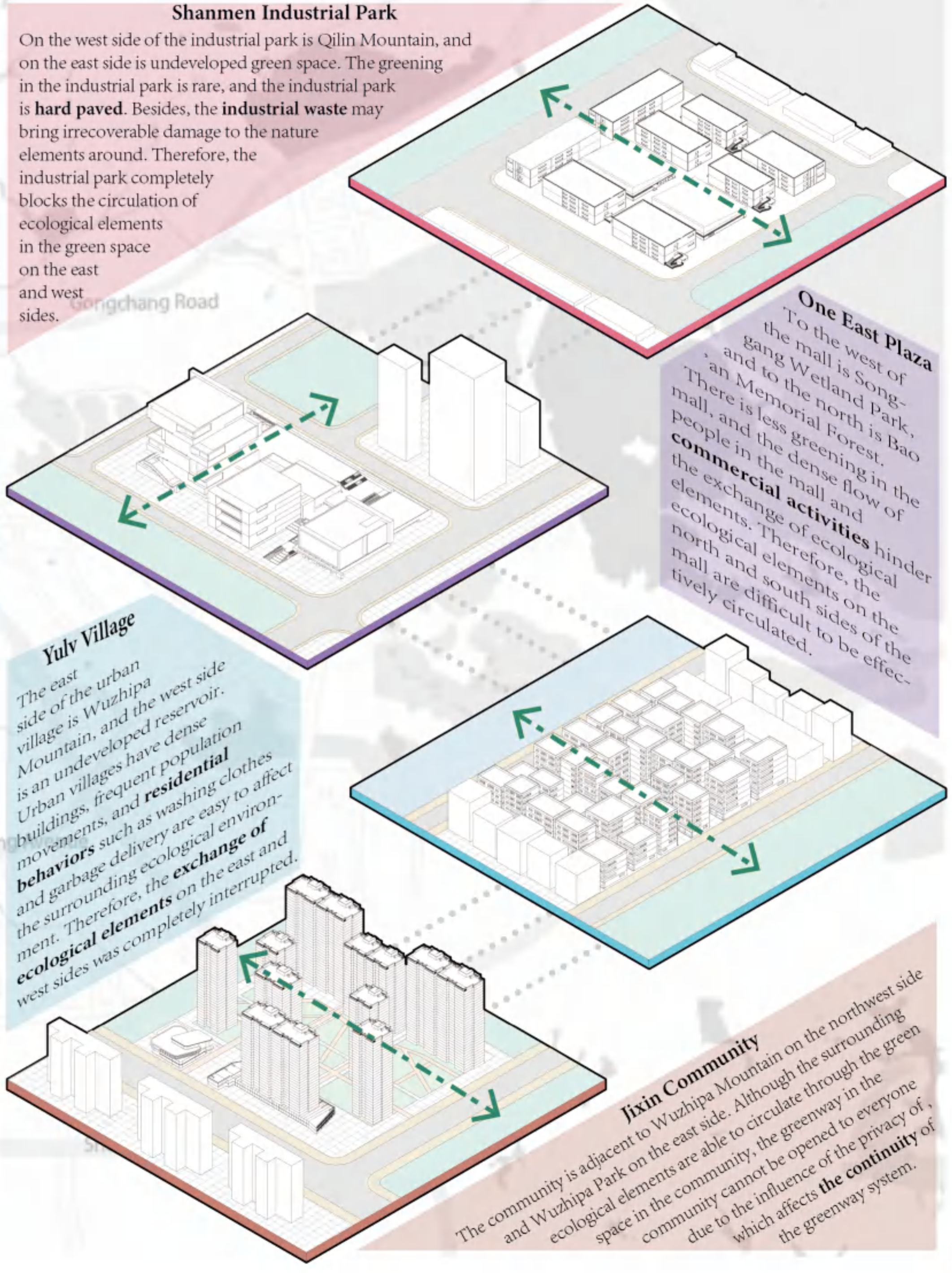
The village covers an area of 8.4 square kilometers with the FAR of about 5.2 and it is adjacent to the Shajing Reservoir in the west. The village has one of the largest underwear factory in China, with more than 20,000 employees at its peak, which basically supports half of the village's economy.

One East Plaza

The total floor area of the mall is 80,000 square meters, and its FAR is about 6.4. It is a comprehensive shopping center integrating commerce, ecological experience, leisure and entertainment with cultural creativity.

Jixin Community

The community covers an area of about 1.4 square kilometers, with a total construction area of about 2.3 square kilometers, the FAR about 1.7, and a green rate of about 47.8%. It is a comprehensive community integrating residential, apartment, high-rise and small high-rise. The community is surrounded by mountains on three sides, enjoying a unique natural environment.



Shanmen Industrial Park

On the west side of the industrial park is Qilin Mountain, and on the east side is undeveloped green space. The greening in the industrial park is rare, and the industrial park is **hard paved**. Besides, the **industrial waste** may bring irrecoverable damage to the nature elements around. Therefore, the industrial park completely blocks the circulation of ecological elements in the green space on the east and west sides.

One East Plaza

To the west of the mall is Songgang Wetland Park, and to the north is Baoan Memorial Forest. There is less greening in the mall, and the dense flow of people in the mall and **commercial activities** hinder the exchange of ecological elements. Therefore, the ecological elements on the north and south sides of the mall are difficult to be effectively circulated.

Yulv Village

The east side of the urban village is Wuzhipa Mountain, and the west side is an undeveloped reservoir. Urban villages have dense buildings, frequent population movements, and **residential behaviors** such as washing clothes and garbage delivery are easy to affect the surrounding ecological environment. Therefore, the **exchange of ecological elements** on the east and west sides was completely interrupted.

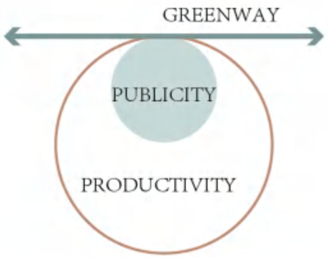
Jixin Community

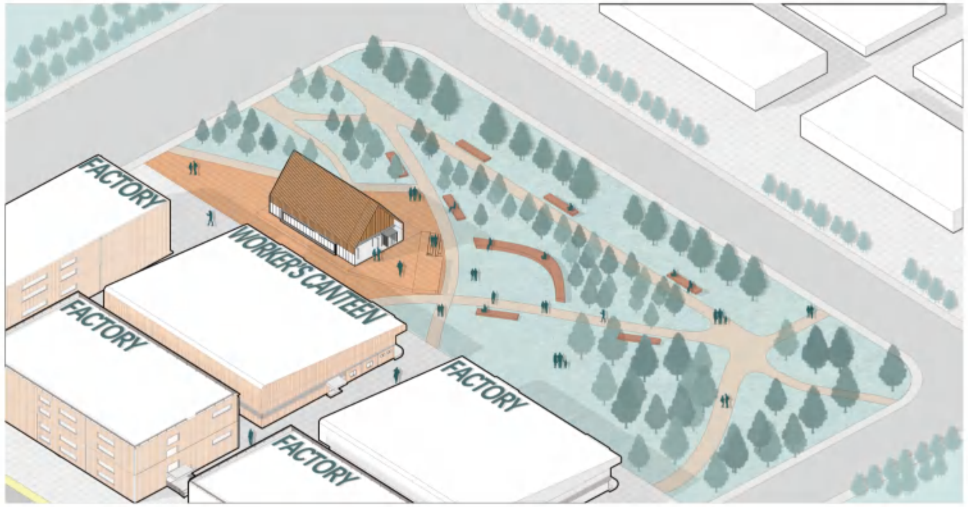
The community is adjacent to Wuzhipa Mountain on the northwest side and Wuzhipa Park on the east side. Although the surrounding ecological elements are able to circulate through the green space in the community, the greenway in the community cannot be opened to everyone due to the influence of the privacy of, which affects **the continuity** of the greenway system.

Industrial Park

Shanmen Industrial Park

This transformation method is suitable for Shanmen Industrial Park, which is representative of **low efficiency and large area** industrial park.
Due to low production capacity, the existing industrial park can be considered for **demolition or integration** into other industrial land. Therefore, part or all of the existing industrial land can be cleared and transformed into **pocket parks**. At the same time, the ecological restoration **exhibition hall** can be used to record the improvement results. Besides, a industrial culture exhibition hall can be used as a transition to connect the industrial park atmosphere, while providing recreation and leisure services, such as a cafe or visitor information center.

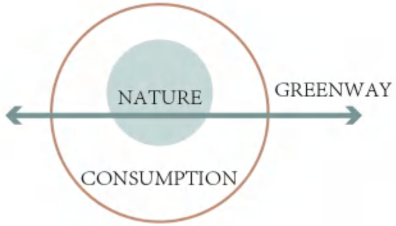


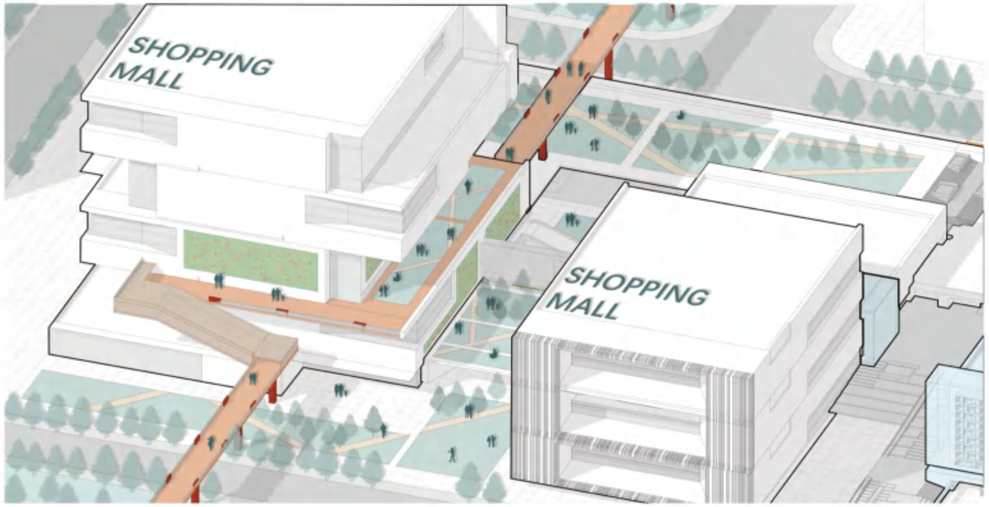


Shopping Mall

One East Plaza

This method is especially suitable for **shopping malls with flexible space and rich forms** represented by One East Plaza.
Taking full advantage of the rich terrain of the mall, the builder can actively build **roof gardens** to extend the green space into the mall. Combined with landscape construction such as path and sculpture, the scattered roof garden can form a variety of forms of public space to provide customers with a better shopping environment.
At the same time, the builders can consider placing the **facade green belt**, especially the seasonal flower wall, to improve the aesthetics of the mall while continuing the green space.

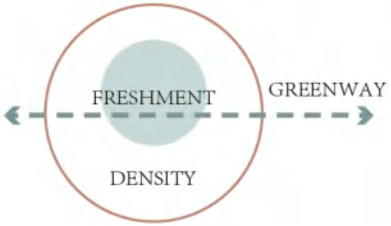


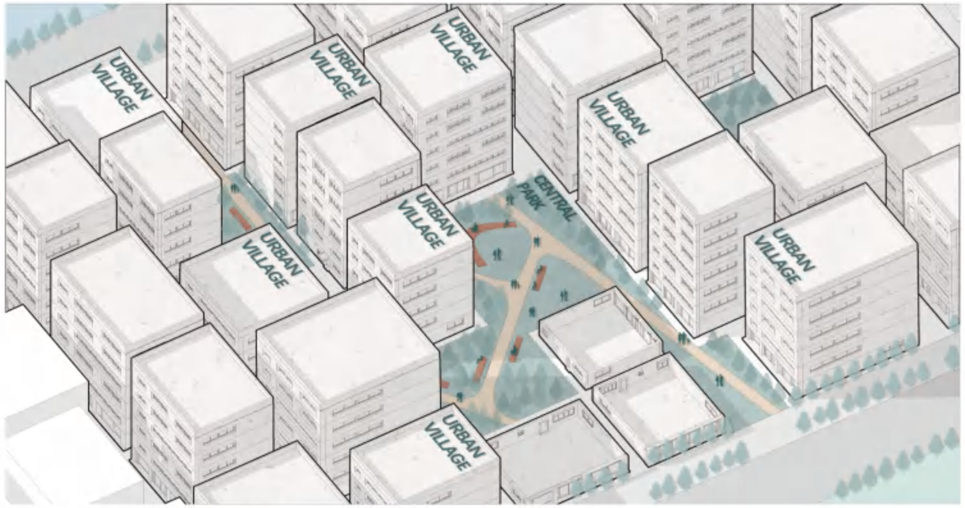


Urban Village

Yulv Village

This transformation method is suitable for the urban villages represented by Yulv Village with **dense buildings and scarce public space**.
Fully excavate the existing open space, integrate the **green space between houses**, and reserve a greenway of sufficient width to connect the fragmented green space between houses.
At the same time, these small green nodes need to be **fully arranged**. For example, they can be combined with setting enough rest seats, building small theaters, and defining drying places, so as to meet the diverse living needs of urban village residents.
In addition, a **community co-governance system** can be established to encourage residents to use the green space between houses to grow vegetables or flowers, and jointly maintain green space.

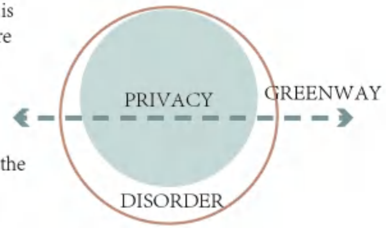


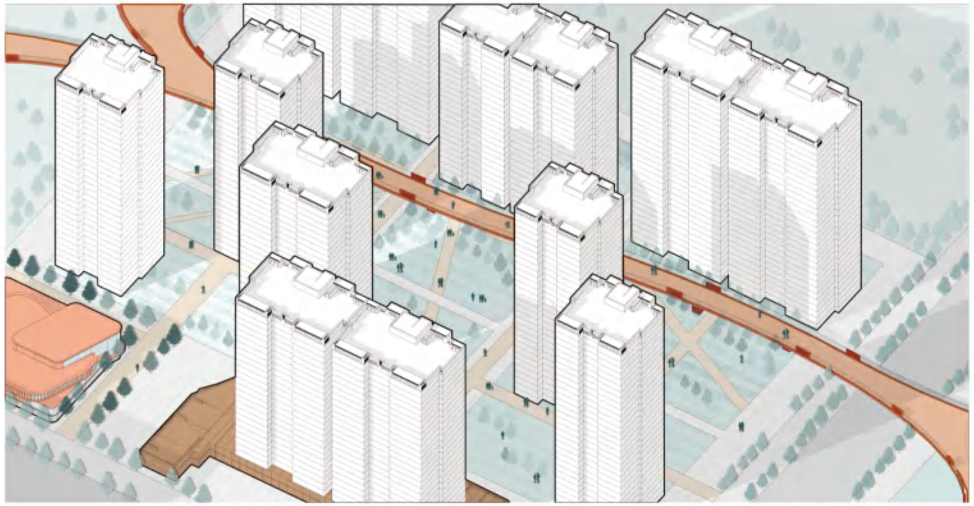


Residential Area

Jixin Community

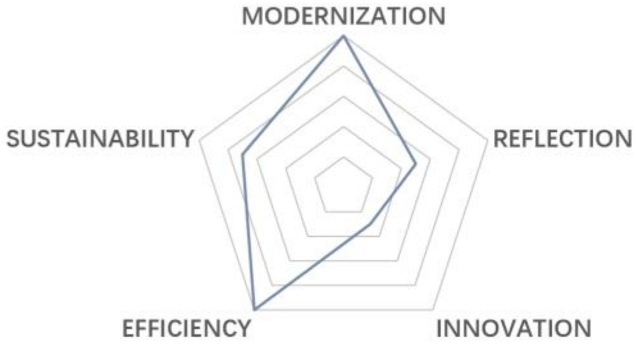
This transformation method is suitable for greenways whose continuity is destroyed by **communities' access control**. In this case, even if there are good green conditions inside the communities, which can connect the green space near the communities and have the prototype of the greenway, the greenway cannot be connected to the public greenway system due to **privacy issues**.
Select the **most lush vegetation and the best greening conditions** in the community to build a **bridge**, and connect the corridor bridge to the urban greenway system.
Due to the dense vegetation, trees can **block the view** to ensure that the privacy of the residents is not violated. At the same time, the ecological atmosphere of trees can be extended to the corridor bridge to ensure the continuity of the greenway atmosphere.





04 OPTIMIZING BIKE SHARING SYSTEMS

A Hybrid Forecasting Model for Temporal and Spatial Demand Analysis



Type: Research Projects
Instructor: Wuyang Hong
Date: 08/2024-11/2024
Location: Chicago, USA
Individual work
Still in progress

Urban planning needs efficiency. As time goes by, the city will have more and more "layers", which will make the city elements richer and more complex. Therefore, the formulation of urban planning cannot only rely on traditional planning theory and empirical judgment, and the emergence of big data technology has greatly improved the efficiency of urban planning

Over the past five decades, bicycle-sharing systems (BSSs) have seen significant evolution and adaptation to accommodate the expansion of urban areas.

This study proposed a new method, aims to predict the future demand for shared-bikes, encompassing both traditional and e-bike models. The research uses the **ARIMA model** to capture temporal trends in 12 months in the year of 2025. Additionally, it employs the **random forest model** to forecast the spatial distribution of this demand in this certain period, ensuring that the location of future bike stations is optimally planned. Furthermore, the research provides comprehensive advice on how to **configure station locations**, as well as determining the number and variety of both traditional and e-bikes required, to cater to the diverse needs and preferences of users across different areas.

1 Introduction

1.1 Background

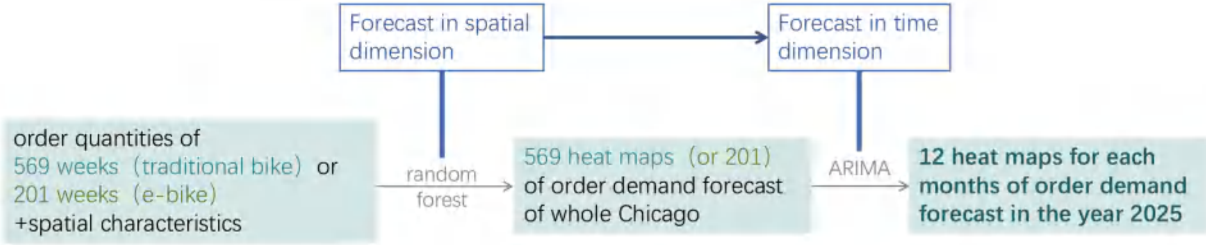
Over the past five decades, bicycle-sharing systems (BSSs) have seen significant evolution and adaptation to accommodate the expansion of urban areas. For a considerable number of people, it has become an integral part of their daily commute. The widespread adoption of bike-sharing systems by urban dwellers has been a testament to their convenience and utility, particularly for short-distance commutes. These systems have become an essential component in facilitating first- and last-mile trips within cities, bridging the gap between public transportation hubs and individual destinations.[1]Moreover, the integration of bike-sharing into the urban transportation infrastructure has provided a multifaceted solution to various urban challenges. By offering a greener mode of transport, bike-sharing systems contribute to the reduction of greenhouse gas emissions.[2] Furthermore, the promotion of cycling as a form of physical activity has positive implications for public health, combating the trend of deteriorating human health associated with modern, sedentary lifestyles.[3]
Despite these benefits, the effectiveness of bike-sharing systems is not without its limitations. A notable issue is the spatial mismatch between demand and supply, which can lead to bicycles being unavailable in certain areas during peak hours or in high-demand locations. This discrepancy not only frustrates users who are unable to access bicycles when and where they need them but also undercuts the system's potential to serve as a reliable transportation option. [4]Consequently, it affects the overall user experience and can deter potential cyclists from using the service. Additionally, this research seeks to shed light on an aspect of bike-sharing systems that has been largely overlooked: the heterogeneity across bike types. The diversity in bicycle designs and functionalities can cater to a wide range of user preferences and needs, yet this variability is under-examined in the context of demand prediction and system optimization. Addressing this oversight is crucial for enhancing the efficiency and user satisfaction of bike-sharing programs.

1.2 Research gap and Purpose

The extant literature on bicycle-sharing demand prediction predominantly focuses on temporal dimensions, with numerous studies attempting to forecast usage patterns over time.[5] While this approach is valuable for understanding and predicting fluctuations in demand throughout the day or across different seasons, it often overlooks the spatial dynamics of bike-sharing systems. Some research has extended its scope to include spatial dimensions, analyzing demand patterns across different areas within a city.[6] However, a notable gap in the current body of knowledge is the lack of integration between these two dimensions.
When predictions are confined to the time dimension alone, they fail to provide actionable insights into the optimal placement of new shared-bike stations. This limitation is significant because the success of a docked bike-sharing system is heavily dependent on the convenience and availability of bicycles at various locations. Conversely, predictions that focus solely on the spatial dimension and ignore the temporal influences on demand may lack the reliability needed to justify the establishment of new stations. Existing research on spatial dimensions typically use datasets with a smaller time span or are based simply on the spatial characteristics of existing sites.[7] Most of these methods fail to observe the influence of time-related factors , such as weather conditions and seasonal variations,on the usage pattern. Therefore, there is a critical need for research that bridges the gap by developing models that integrate both temporal and spatial dimensions to provide more comprehensive and reliable predictions for bike-sharing demand.

1.3 Research Framework

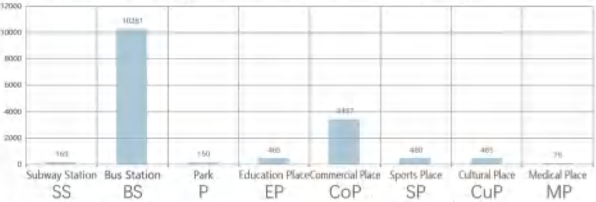
This study proposed a new method, aims to predict the future demand for shared-bikes, encompassing both traditional and e-bike models, in both time and spatial dimensions. The research will use the ARIMA model to capture temporal trends in 12 months in the year of 2025. Additionally, it employs the random forest model to forecast the spatial distribution of this demand in this certain period, ensuring that the location of future bike stations is optimally planned. Furthermore, the research provides comprehensive advice on how to configure station locations, as well as determining the number and variety of both traditional and e-bikes required, to cater to the diverse needs and preferences of users across different areas.

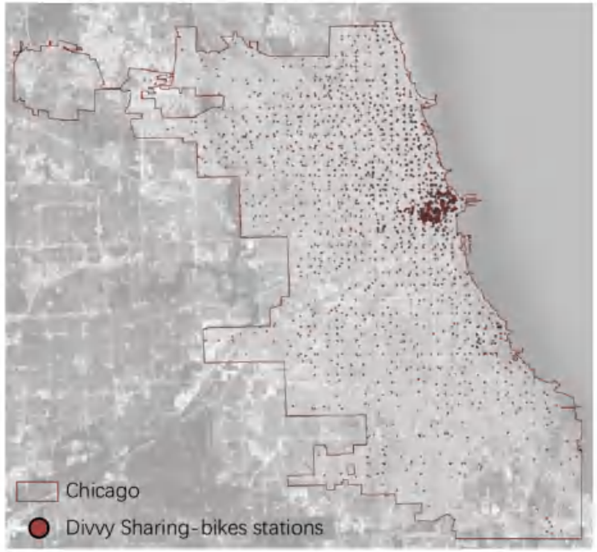


2 Materials

2.1 Study Area

Chicago's Divvy system belongs to the latest generation of bicycle-sharing systems that have benefited from recent advances in technology such as solar-powered docking stations. The latest location of Divvy stations is presented in Fig 1. Chicago is the third most populous city in the US and hosts millions of visitors every year.
The Chicago downtown area is the second largest commercial business district in the U.S. According to the NHTS (2009), bicycle trips account for 1.7% of total trips in Chicago, while 80.6% of trips are made by private vehicles. About 44.8% of trips are less than 2 miles; within the trips less than 2 miles, the private-vehicles share drops to 66.3%, while the share for bicycle mode increases to 2.9%. The statistics clearly indicate the scope for shifting some of the urban core trips from private vehicles to shared bicycling.





2.2 Data Source

Trip data of Divvy system can be downloaded from the official website. Moreover, every trip record contains detailed information including the time, latitude and longitude of start (end) station, user attributes (age and gender only for subscribers). In particular, the ride-able type occurred in the record since August, 2020, as a result that electric sharing bikes (e-bikes) were added into the system.

ride_id	rideable_type	started_time	ended_time	start_station_name	start_station_id	end_station_name	end_station_id	start_lat	start_lng	end_lat	end_lng	member_casual
09DF0C5B0C0059B1	classic_bike	2024/1/1 0:08	2024/1/1 0:14	Wells St & Hubbard St	TA1307000151	Wells St & Walton St	TA1306000011	41.889906	-87.634266	41.89993001	-87.63443007	member
5BAAFDE84176D405	electric_bike	45292.00764	45292.01181	California Ave & Francis Pl	13259	Milwaukee Ave & Wabansia Ave	13243	41.9186512	-87.6973379	41.912616	-87.681391	casual

Other data utilized in this study encompasses a comprehensive array of demographic, social, and geographical datasets for the city of Chicago.Land use data, which categorizes the types of activities and developments across the city, is integrated to understand spatial distribution patterns. To capture the urban infrastructure and points of interest (POIs), the research have included data on public transit stations (bus and metro), parks, educational institutions, sports facilities, commercial areas, and medical facilities. This POI data is crucial for identifying key locations that may influence the usage patterns of bike-sharing systems. Furthermore, the road network data offers a detailed map of the city's streets, which is essential for analyzing connectivity and travel routes, and the roads has been classified into 20 categories in this dataset. All these datasets are classified as open data, which means they are freely available to the public and can be accessed and used by anyone.
The study uses QGiS (ver. 3.38.0)to process the spatial data, such as road network data, and uses EPSG:4326-WGS 84 as the unifed coordinate system. Meanwhile, the study uses PyCharm Community Edition 2023.3.5 to process other data, such as trip data.

2.3 Data Preparation

Firstly, the study divides the Chicago into 4657 grids of 500m*500m.

Y-

Order Quantity - The research selects all the available trip data, and calculates the quantities of trip order weeks by weeks, which provides 569 monthly trip order data of traditional bike and 201 of e-bike. The number of trips within each grid is represented by the number of trips departing in that grid during the study time. One week (seven days) was chosen as the minimum time unit for the study.

X-

POI - Count the number of different types of POIs in each grid.
Road Network(RN)- ① Divide the road network into 20 grades, and set steps and others to 0.
②Use a grid to cut the road network into small segments, and calculate the length of each segment in each grid.
③ Calculate road weight per grid =*ROAD1* length * *ROAD1* grade + *ROAD2* length * *ROAD 2* grade....
Month(M) - The month in which the first day of the week falls.
Day of Year(DoY) - The day of the year on which the first day of the week falls.

For each dependent and independent variable in the grid, the research does normalization processing , and the formula is as follows:

$X^* = (X - min) / (max - min)$

Where X^* represents the normalized value of each variable in each grid; X denotes the original value of each variable, ***min*** signifies the minimum value of each variable, and ***max*** denotes the maximum value of each variable.

2.4 Methods

Random Forest

Random forests are built using a technique called "bagging" (bootstrap aggregating) with classification and regression trees (CART). In this process, multiple bootstrap samples are drawn from the original training data, and CART models are created for each sample. The results from all the trees are then averaged, which helps reduce variance and prevents overfitting.
There are two main tools used to interpret random forest models: variable importance and partial dependence plots. Variable importance measures how much each variable contributes to the model's predictions. In tree-based models, it's typically determined by looking at the improvement in prediction accuracy when a split is made based on a variable. For random forests, the variable importance is averaged across all the trees.
Partial dependence plots help visualize how an independent variable affects the dependent variable. These plots show the average model output at different values of an independent variable, while holding the other variables constant. This process is repeated for a range of values of the variable of interest, creating a plot that shows its effect (Friedman, 2001).

Training

①Using default parameters.
②Grids with no trips are eliminated in the training set.
③The sliding window method is used to determine the window size, and set the step size to one week.
When the window is within one year, the larger the window is, the smaller the NMAE is.
The the smallest NMAE, which is 0.14, occurs in the week from May 1, 2017 to April 30, 2018.

ARIMA

The ARIMA model, standing for AutoRegressive Integrated Moving Average, combines three main components for time series forecasting: autoregression (AR), differencing (I) to make the series stationary, and moving average (MA). Mathematically, an ARIMA model is denoted as ARIMA (p, d, q), where "p"is the order of the AR terms, "d" is the degree of differencing needed to make the series stationary, and "q"is the order of the MA terms. This model predicts future values based on past values (AR), the differences between past values (I), and forecast errors (MA).
The study randomly selected 40 grids, and calculated their best parameters as experimental parameters, and finally chose p=2, d=0, q=2.

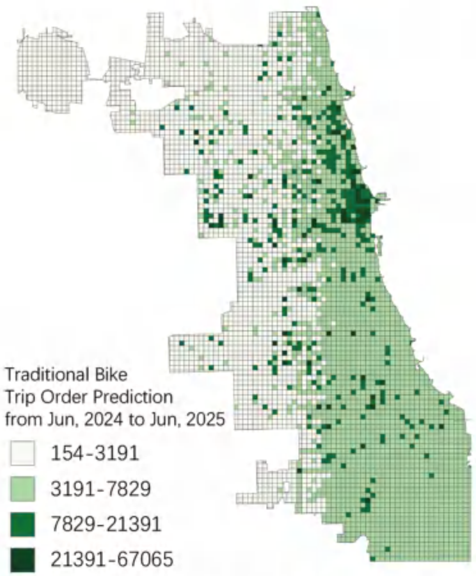
3 Results and Discussions

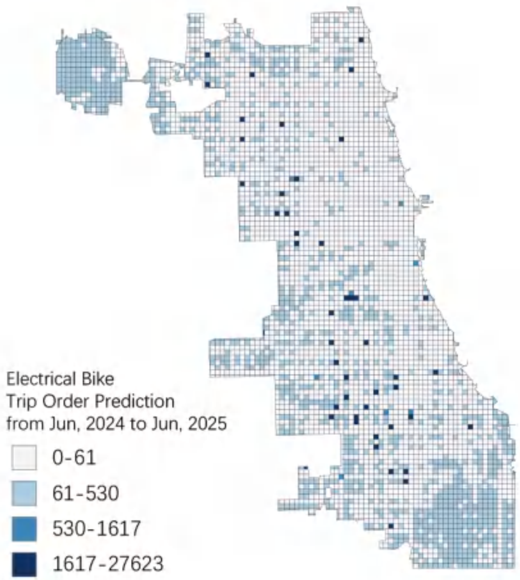
3.1 Predicting Results in Spatial Dimension

Feature Importance for Traditional Bike										
POI								RN	M	DoY
SS	BS	P	EP	CoP	SP	CuP	MP			
1.8	5.3	0.2	0.8	28.3	0.2	2.1	1.2	37.5	3.3	19.4

Feature Importance for Electrical Bike										
POI								RN	M	DoY
SS	BS	P	EP	CoP	SP	CuP	MP			
0.9	7.4	0.1	1.7	31.6	0.4	1.6	1.4	35.8	4.5	14.9

3.2 Predicting Results in Time Dimension





*The natural breakpoint grading method was used for classification.

3.3 Discussions

In general, orders for traditional bikes will still be larger than those for electric bikes. While traditional bikes see more orders near the city center, areas where e-bikes are heavily used are scattered throughout the city and outside the city center. This may be due to the dense building elements in the downtown area, and the short travel distance of people, so there is a lower demand for electric vehicles. By contrast, The buildings in the suburbs of the city are scattered, and the average travel distance of people is relatively long, which requires the help of electric vehicles.
According to the prediction results, it should be considered to appropriately increase or reduce the existing stations, and adjust the station vehicle ratio. More traditional bicycle stations should be set up in the city center, and more electric vehicles should be placed in the stations outside the city center, even if the number of stations can be reduced accordingly.
The popularity of both types of bikes is heavily influenced by the road network, bus stations and commercial facilities. Therefore, when choosing the location of a bicycle station, the bicycle supplier should try to locate the station close to the roads which are suitable for cycling, bus station or shopping mall , or consider coupling the station with the above facilities.

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